

Central Valley Regional Water Quality Control Board

22 August 2018

Chandra Johannesson
East Bay Municipal Utility District
375 11th Street
Oakland, CA 94607

CERTIFIED MAIL
91 7199 9991 7036 7006 7310

NOTICE OF APPLICABILITY
GENERAL WASTE DISCHARGE REQUIREMENTS FOR
SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS
ORDER WQ 2014-0153-DWQ
FOR
EAST BAY MUNICIPAL UTILITY DISTRICT
PARDEE RESERVOIR RECREATION AREA
WASTEWATER TREATMENT PLANT
AMADOR COUNTY

The East Bay Municipal Utility District (EBMUD, hereafter "Discharger") submitted a Report of Waste Discharge (RWD) dated 28 August 2017 describing the Pardee Reservoir Recreation Area Wastewater Treatment Plant (WWTP) in Amador County. Based on the information provided, the wastewater treatment system and discharge is consistent with the requirements of the State Water Resources Control Board (State Water Board) *General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems*, Order WQ 2014-0153-DWQ (General Order). This Notice of Applicability (NOA) provides notice that the General Order is applicable to the site as described below. You are hereby assigned Order WQ 2014-0153-DWQ-R5287 for the discharge. A copy of the General Order is enclosed and also available at:

http://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

You should familiarize yourself with the entire General Order and its attachments, which describe mandatory discharge and monitoring requirements. The General Order contains operational and reporting requirements by wastewater system type. Sampling, monitoring, and reporting requirements applicable to your treatment and disposal methods must be completed in accordance with the appropriate treatment system sections of the General Order and the attached Monitoring and Reporting Program (MRP) 2014-0153-DWQ-R5287. The Discharger is responsible for all the applicable requirements that exist in the General Order and this NOA.

REGULATORY BACKGROUND

WDRs Order R5-01-270, adopted by the Central Valley Regional Water Quality Control Board (Central Valley Water Board) on 7 December 2001, prescribes requirements for the wastewater treatment system and allows a monthly average dry weather inflow of 11,500 gallons per day (gpd). WDRs Order R5-01-270 is proposed to be rescinded at the **4/5 October 2018** meeting of the Central Valley Water Board. Effective upon rescission of Order R5-01-270, the discharge described in this NOA shall be regulated pursuant to the General Order.

FACILITY AND DISCHARGE DESCRIPTION

The WWTP is located at 4900 Stony Creek Road, lone in Amador County as shown on Attachment A, which is incorporated herein. The Assessor's Parcel Number for the WWTP is 012-080-011 in Section 15, T5N, R10E, MDB&M. The facility is at the north end of the Pardee Reservoir. The recreation area lot size is 245.9 acres.

The Discharger owns and operates the WWTP, which provides wastewater treatment for Pardee Reservoir and the recreation area. The recreation area's collection system and the WWTP were originally built in the 1980s. The recreation area includes five mobile homes for the concessionaire's management and staff, a maintenance yard, a recreational vehicle park with a capacity of 56 sanitary sewer hook-ups, and a hundred campsites. The WWTP treats and discharges wastewater generated at the recreation area except from the onsite campsites as they are equipped with portable restrooms.

The recreation area is open to the public from February through October with year-long usage for concessionaire and maintenance staff. The daily average flow during the peak summer season is about 3,700 gallons per day (gpd), with the winter flow at 600 gpd. The recreation area's water supply is from Pardee Reservoir, which collects water from the Mokelumne River with low salinity (TDS 34 mg/L based a sample collected in July 2017).

The WWTP consists of two concrete-lined aerated treatment/storage ponds operated in series (East and West Ponds), a chlorine contact chamber and a six-acre land application area (LAA). The combined capacity of East Pond and West Pond is approximately 13.8 acre feet based on two-feet of free board. In late 2000, the ponds were lined with four to eight-inch thick reinforced concrete due to seepage and surfacing of water downgradient of the then unlined pond (north end).

The wastewater gravity flows through the recreation area's sanitary sewer collection system. The influent, measured through a weir style flow meter, flows directly into the East Pond and then through a set weir into West Pond. The West Pond is equipped with a pump which pumps the wastewater from the West Pond to the 2,314-gallon L-shaped chlorine contact chamber where sodium hypochlorite is added for disinfection. Secondary disinfected effluent is applied to the LAA for disposal. A tailwater return ditch system is located at the base of the LAA (upper right corner of the West Pond). The tailwater is pumped back into the West Pond. Rainwater and groundwater seepage along the ponds drains offsite through a French drain system. The WWTP is connected to the EBMUD's District-wide Supervisory Control and Data Acquisition System.

A site plan and a process schematic are shown on Attachment B and C, respectively, which are incorporated herein.

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A site plan and a process schematic are shown on Attachment B and C, respectively, which are incorporated herein.

The table below is a summary of effluent quality based on two monitoring reports.

Constituent (mg/L)	June 2016 Monitoring Report	July 2017 Monitoring Report
Total Dissolved Solid	510	600
Sodium	106	96
Chloride	120	100
Biochemical Oxygen Demand (BOD ₅)	5.1	7.1
Total Kjeldahl Nitrogen	5.6	7.8
Nitrate as Nitrogen	<0.02	0.03
Formaldehyde	0.03	0.04
Zinc	0.01	0.02
Phenol	<0.03	<0.01

Three groundwater monitoring wells MW-1, MW-2 and MW-3 were installed in 2003 near the ponds as shown on Attachment B. Based on measurements collected in July 2017, the depths to groundwater range from 6 to 48 feet below ground surface, and the groundwater downgradient direction was to the northeast with an approximate horizontal gradient of 0.02 feet per foot. In most years, groundwater elevations in MW-1 were higher than in MW-2 and MW-3. However, during some dry years, MW-1 was typically downgradient of the compliance wells. A summary of historical groundwater monitoring data is presented in the table below based on semi-annual data collected from February 2012 through July 2017.

Constituent	Up-Gradient Well	Down-Gradient Well		Concentration Protective of Beneficial Uses
	MW-1 (Mean)	MW-2 (Mean)	MW-3 (Mean)	
TDS (mg/L)	300	380	440	450 ¹ to 1,500 ²
Sodium (mg/L)	11	26	28	69 ¹
Chloride (mg/L)	8.9	55	53	106 ¹ - 250 ⁴
Nitrate Nitrogen (mg/L)	14	6.5	5.7	10 ³
Zinc (µg/L)	9.7	22	14	5,000 ⁴
Formaldehyde (µg/L)	<10	<10	<10	100 ⁵
Phenols (mg/L)	<0.03	<0.02	<0.03	4.2 ⁵

¹ Lowest agricultural water quality goal.

² Short-term Secondary Maximum Contaminant Level.

³ Primary Maximum Contaminant Level.

⁴ Secondary Maximum Contaminant Level.

⁵ California Notification Levels for drinking water.

SITE-SPECIFIC REQUIREMENTS AND EFFLUENT LIMITS

The wastewater treatment operator must be familiar with the requirements contained in the General Order, this NOA, and the MRP.

Note that the General Order contains prohibitions and specifications that apply to all wastewater treatment systems as well as those that only apply to specific treatment and/or disposal systems. The specific requirements for your treatment system are summarized below.

Requirements by Wastewater System Type, Section B of General Order

A. Prohibitions

This section applies to all discharges.

B. Requirements by Wastewater System Type

B.1 All Wastewater Systems

B.1.a Inflow to the WWTP shall not exceed **11,500 gpd** as an average dry weather flow defined as the total flow for the months of July through September divided by 92 days.

B.1.l Wastewater system setbacks.

Equipment or Activity	Domestic Well	Flowing Stream	Ephemeral Stream Drainage	Property Line	Lake or Reservoir
Septic Tank, Treatment System, & Collection System ¹	150 ft.	50 ft.	50 ft.	5 ft.	200 ft.
LAND APPLICATION AREA REQUIREMENTS					
LAA (disinfected sec-2.2 or sec-23 recycled water)	100 ft.	50 ft.	50 ft.	50 ft.	200 ft.
WASTEWATER STORAGE AND/OR TREATMENT PONDS					
Impoundment (disinfected secondary recycled water) ²	150 ft.	150 ft.	150 ft.	50 ft.	200 ft.

¹ Reference setbacks from "Septic Tank, Aerobic Treatment Unit, Treatment System, or Collection System" in Table 3 of General Order.

² Reference setbacks from "Wastewater Storage and/or Treatment Ponds" in Table 3 of General Order.

B.5 Pond Systems

The WWTP utilizes an evaporation/percolation pond; therefore this section applies in its entirety.

B.7 Land Application and/or Recycled Water Systems

The WWTP utilizes a LAA; therefore, this section applies in its entirety. In addition, wastewater shall only be applied to the LAA from **April through October** each year.

Effluent Limitations, Section D of General Order

This section applies in its entirety to the WWTP and shall include the following site specific limitations.

Effluent Limitations

The following limit apply to the effluent prior to discharge to the LAA.

Constituent	Units	30-Day Average	Daily Maximum
BOD ₅	mg/L	40	80
Total Coliform Organisms	MPN/100mL	23	240

MONITORING AND REPORTING PROGRAM

The Discharger shall comply with MRP 2014-0153-DWQ-R5287, which is incorporated herein.

ENFORCEMENT

Please review this NOA carefully to ensure that it completely and accurately reflects the discharge. Discharge of wastes other than those described in this NOA is prohibited. Prior to allowing changes to the wastewater strength or generation rate, or to the method of waste disposal, you must contact the Central Valley Regional Water Board to determine if submittal of an RWD is required.

The Discharger generates the waste subject to the terms and conditions of the General Order and maintains exclusive control over the discharge. As such, the Discharger is primarily responsible for compliance with this NOA, MRP, and General Order, with all attachments. Failure to comply with the requirements in the General Order or this NOA could result in an enforcement action as authorized by provisions of the California Water Code.

ANNUAL FEES

Staff has determined the discharge is a threat to water quality and complexity rating of 3-B. The annual fee corresponding to a threat to water quality and complexity of 3-B is currently \$4,699; however, because the permitted flow is less than 50,000 gpd, the discharge qualifies for the 50-percent fee discount. Therefore, the annual fee for this discharge is currently \$2,350. The fee is due and payable on an annual basis until coverage under the General Order is formally rescinded. Please note that the annual fees are reviewed each year and may change. If the wastewater discharge ceases, you must provide written notice so that we can terminate coverage under the General Order and no longer bill you.

DOCUMENT SUBMITTAL

All monitoring reports and other correspondence should be converted to searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50 MB should be emailed to:

centralvalleysacramento@waterboards.ca.gov.

To ensure that your submittal is routed to the appropriate staff person, the following information should be included in the body of the email or any documentation submitted to the mailing address for this office:

Facility Name: East Bay Municipal Utility District, Pardee Reservoir Recreation Area Wastewater Treatment Plant WWTP, Amador County		
Program: Non-15 Compliance	Order: 2014-0153-DWQ-R5287	CIWQS Place ID: 247616

Documents that are 50 MB or larger should be transferred to a CD, DVD, or flash drive and mailed to:

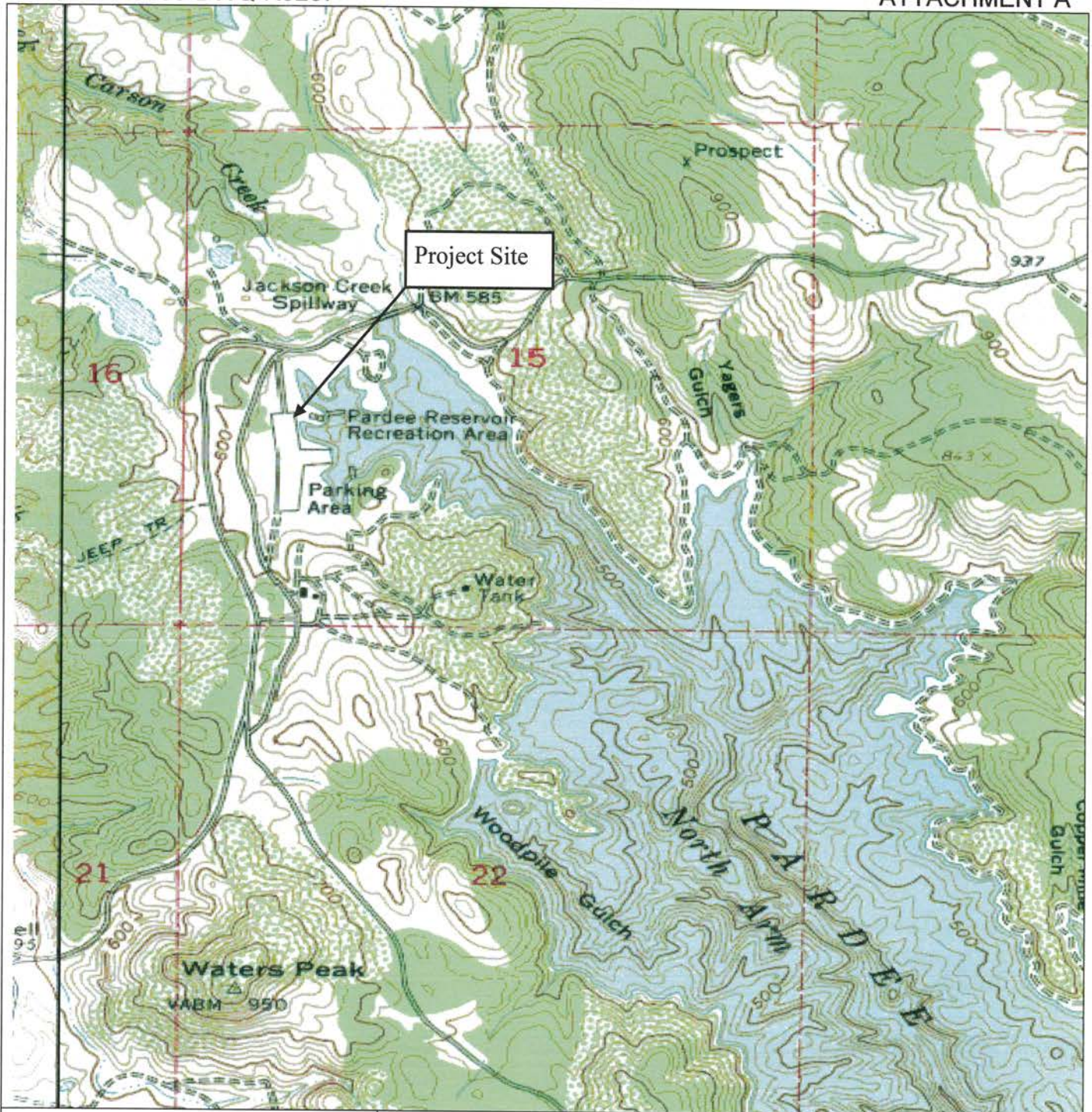
Central Valley Regional Water Quality Control Board
ECM Mailroom
11020 Sun Center Drive, Suite 200
Rancho Cordova, CA 95670

Now that the Notice of Applicability has been issued, the Board's Compliance and Enforcement section will take over management of your case. Kenny Croyle is your new point of contact for any questions about the General Order. If you find it necessary to make a change to your permitted operations, Kenny will direct you to the appropriate Permitting staff. You may contact Kenny at (916) 464-4676 or at kcroyle@waterboards.ca.gov.


For Patrick Pulupa
Executive Officer

enc: Water Quality Order WQ 2014-0153-DWQ
Monitoring and Reporting Program 2014-0153-DWQ-R5287
Attachment A, Site Location Map
Attachment B, Site Plan
Attachment C, Process Schematic
MRP Transmittal Sheet

cc w/out enc: Timothy O'Brien, State Water Resources Control Board, Sacramento
Michael Israel, Amador County Environmental Health Department, Jackson



Drawing Reference:

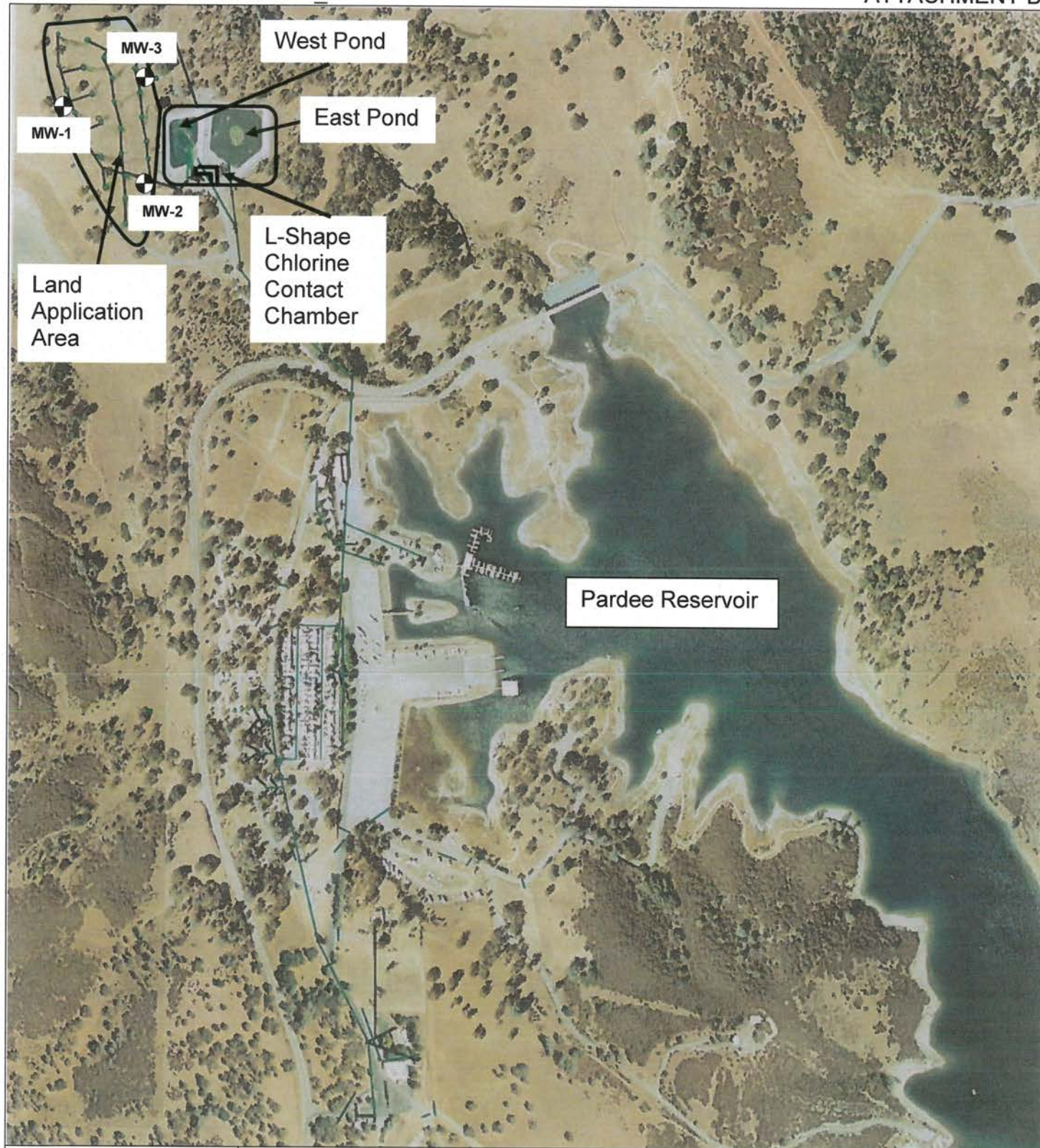
U.S.G.S TOPOGRAPHIC MAP
7.5 MINUTE QUADRANGLE

SITE LOCATION MAP

EAST BAY MUNICIPAL UTILITY DISTRICT
PARDEE RECREATION AREA WWTP
AMADOR COUNTY



approx. scale
1 in. = 2,400 ft.



Drawing Reference:

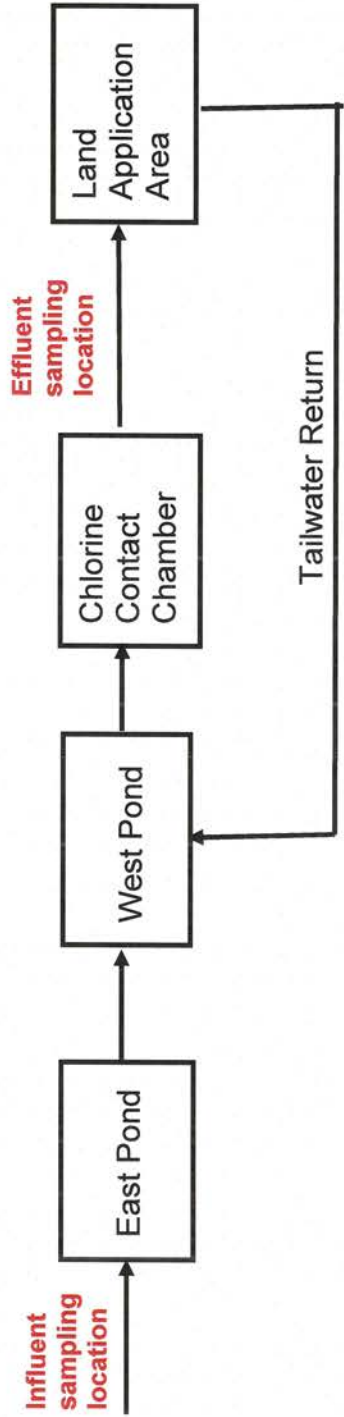
Report of Waste
Discharge,
August 2017

SITE PLAN

EAST BAY MUNICIPAL UTILITY DISTRICT
PARDEE RESERVOIR RECREATION AREA
WASTEWATER TREATMENT PLANT
CALAVERAS COUNTY



Not Scaled



Drawing Reference:
RWD, August 2017

PROCESS SCHEMATIC
EAST BAY MUNICIPAL UTILITY DISTRICT
PARDEE RESERVOIR RECREATION AREA
WASTEWATER TREATMENT PLANT
CALAVERAS COUNTY

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM WQ 2014-0153-DWQ-R5287

FOR
EAST BAY MUNICIPAL UTILITY DISTRICT
PARDEE RESERVOIR RECREATION AREA
WASTEWATER TREATMENT PLANT
AMADOR COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring influent, ponds, effluent, land application area, groundwater, sludge and water supply. This MRP is issued pursuant to Water Code Section 13267. The East Bay Municipal Utility District (Discharger) shall not implement any changes to this MRP unless and until another revision is issued by the Executive Officer.

The Discharger operates the wastewater system that is subject to the Notice of Applicability (NOA) of Water Quality Order 2014-0153-DWQ-R5287. The reports are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to Water Code section 13267, the Discharger shall implement this MRP and shall submit the monitoring reports described herein.

All wastewater samples should be representative of the volume and nature of the discharge. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. Wastewater flow monitoring shall be conducted continuously using a flow meter and shall be reported in cumulative gallons per day.

Field test instruments (such as pH and dissolved oxygen) may be used if:

1. The operator is trained in the proper use of the instrument;
2. The instruments are field calibrated prior to each use;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are submitted as described in the "Reporting" section of this MRP.

Analytical procedures shall comply with the methods and holding times specified in the following: Methods for Organic Chemical Analysis of Municipal and Industrial Wastewater (EPA); Test Methods for Evaluating Solid Waste (EPA); Methods for Chemical Analysis of Water and Wastes (EPA); Methods for Determination of Inorganic Substances in Environmental Samples (EPA); Standard Methods for the Examination of Water and Wastewater (APHA/AWWA/WEF); and Soil, Plant and Water Reference Methods for the Western Region (WREP 125). Approved editions shall be those that are approved for use by the United States Environmental Protection Agency or the California Department of Public Health's Environmental Laboratory Accreditation Program. The Discharger may propose alternative methods for approval by the Executive Officer. Where technically feasible,

laboratory reporting limits shall be lower than the applicable water quality objectives for the constituents to be analyzed.

INFLUENT MONITORING

Influent monitoring shall be performed at the headworks. Influent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Meter	Daily	Monthly
20°C BOD ₅ ¹	mg/L	Grab	Monthly	Monthly

¹ 5-day Biochemical Oxygen Demand.

POND MONITORING

Samples shall be collected from an established sampling station. Freeboard shall be measured vertically from the surface of the water to the lowest elevation of the surrounding berm and shall be measured to the nearest 0.1 feet. Monitoring of each pond shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Dissolved Oxygen ¹	mg/ L	Grab	Weekly	Monthly
pH	Standard units	Grab	Weekly	Monthly
Freeboard	0.1 feet	Measurement	Weekly	Monthly
Odors	--	Observation	Weekly	Monthly

¹ If the pond is too low to take a dissolved oxygen reading, then this shall be noted on the monthly monitoring report.

EFFLUENT MONITORING

Effluent samples shall be collected from April through October. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹	<u>Reporting Frequency</u>
Biological Oxygen Demand ⁵	mg/L	Grab	Weekly	Monthly
Total Coliform Organisms	MPN /100 mL	Grab	Weekly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
pH	Std units	Grab	Monthly	Monthly
Formaldehyde	mg/L	Grab	Monthly, May-Sept.	Monthly, May-Sept.
Zinc	mg/L	Grab	Monthly, May-Sept.	Monthly, May-Sept.
Phenol	mg/L	Grab	Monthly, May-Sept.	Monthly, May-Sept.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹	<u>Reporting Frequency</u>
Standard Minerals ²	mg/L	Grab	Annually	Annually

¹ Samples shall be collected between the months of April and October, unless described otherwise.

² Standard Minerals shall include, at a minimum, the following elements/compounds: boron, iron, manganese, calcium, magnesium, potassium, sodium, chloride, sulfate, total alkalinity (including alkalinity series), and hardness.

LAND APPLICATION AREA MONITORING

Monitoring of the land application area shall be conducted daily when wastewater is applied (April through October), and the results shall be included in the monthly monitoring report. Evidence of erosion, field saturation, runoff, or the presence of nuisance conditions shall be noted in the report. The tailwater collection ditch and valves shall be checked for leaks and overflows. Effluent monitoring results shall be used to calculate loading rates at the application area. Monitoring of the land application area shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gallons	Continuous	Daily	Monthly
Rainfall	Inches	Measurement	Daily	Monthly
Acreage Applied	Acres	Calculated	Daily	Monthly
Application Rate	gal/acre•day	Calculated	Daily	Monthly
BOD Loading Rate	lbs/acre•day	Calculated	Monthly	Monthly
Total Nitrogen Loading Rate	lbs/ac/month	Calculated	Monthly	Monthly
TDS Loading Rate	lbs/ac/month	Calculated	Monthly	Monthly

BOD denotes Biological Oxygen Demand. TN denotes Total Nitrogen. TDS denotes Total Dissolved Solids.

GROUNDWATER MONITORING

Groundwater samples shall be collected from each groundwater monitoring well in accordance with an approved groundwater monitoring workplan. Prior to sampling, depth to groundwater shall be measured to the nearest 0.01 feet. Water table elevations shall be calculated and used to determine groundwater gradient and flow direction. Samples shall be collected and analyzed using approved EPA methods or other methods approved by the Central Valley Water Board. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Depth to Groundwater	± 0.01 feet	Measurement	Semi-annually
Groundwater Elevation	± 0.01 feet	Calculated	Semi-annually
Gradient	feet/feet	Calculated	Semi-annually
Gradient Direction	degrees	Calculated	Semi-annually
pH	pH units	Grab	Semi-annually

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Chloride	mg/L	Grab	Semi-annually
Sodium	mg/L	Grab	Semi-annually
Total Coliform Organisms	MPN/100 mL	Grab	Semi-annually
Total Dissolved Solids	mg/L	Grab	Semi-annually
Nitrate-N	mg/L	Grab	Semi-annually
Total Trihalomethanes	µg/L	Grab	Semi-annually
Zinc	mg/L	Grab	Semi-annually
Total Phenols	mg/L	Grab	Semi-annually
Formaldehyde	mg/L	Grab	Semi-annually
Standard Minerals ¹	mg/L	Grab	Annually

¹ Standard Minerals shall include the following compounds: boron, dissolved iron, dissolved manganese, calcium, magnesium, potassium, sulfate, total alkalinity (including alkalinity series), and hardness.

SOLID WASTE AND SLUDGE MONITORING

A log shall be kept of solid waste (grits and screenings) and sludge quantities generated and of handling and disposal activities, and shall be submitted as part of the monthly monitoring reports.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Total Dissolved Solids	mg/L	Annually
pH	pH units	Annually
Standard Minerals ¹	mg/L	Annually

¹ Standard Minerals shall include, at a minimum, the following constituents: boron, iron, manganese, calcium, magnesium, potassium, sulfate, chloride, sodium, total alkalinity (including alkalinity series), and hardness.

REPORTING

All regulatory documents, submissions, materials, data, monitoring reports, and correspondence should be converted to a searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50MB should be emailed to:

centralvalleysacramento@waterboards.ca.gov

Documents that are 50 MB or larger should be transferred to a CD, DVD, or flash drive and mailed to the following address:

Central Valley Regional Water Quality Control Board
ECM Mailroom
11020 Sun Center Drive, Suite 200
Rancho Cordova, California 95670

To ensure that your submittals are routed to the appropriate staff, the following information block should be included in any correspondence used to transmit documents to this office:

Facility Name: East Bay Municipal Utility District, Pardee Reservoir Recreation Area Wastewater Treatment Plant WWTP, Amador County		
Program: Non-15 Compliance	Order: 2014-0153-DWQ-R5287	CIWQS Place ID: 247616

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the MRP shall be reported in the next scheduled monitoring report.

In addition to the requirements of Standard Provision C.3, monitoring information shall include the method detection limit (MDL) and the Reporting limit (RL) or practical quantitation limit (PQL). If the regulatory limit for a given constituent is less than the RL (or PQL), then any analytical results for that constituent that are below the RL (or PQL) but above the MDL shall be reported and flagged as estimated.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a California Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Daily, weekly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, pond, effluent, and land application area monitoring;
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format; and
3. If requested by staff, copies of laboratory analytical report(s).

B. Semi-annual Monitoring Reports

In addition to the monthly monitoring reports, the Discharger shall establish quarterly and semi-annual sampling schedules for groundwater monitoring such that samples are obtained approximately every three and six months. Semi-annual monitoring reports shall be submitted to the Board by the **1st day of February and August**. The semi-annual reports shall include the following:

1. Results of groundwater monitoring;
2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for groundwater monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater and method of sampling;
3. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any;
4. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);
5. Summary data tables of historical and current groundwater table elevations and analytical results;
6. A comparison of monitoring data to the groundwater limitations and an explanation of any violation of those requirements; and
7. Copies of laboratory analytical report(s) for groundwater monitoring.

C. Annual Report

In addition to the monthly and semi-annual monitoring reports, an Annual Report shall be prepared. The Annual Report shall be submitted to the Regional Board by **1 February** each year. The Annual Report shall include the following:

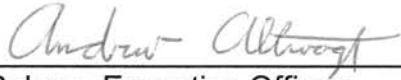
1. If requested by staff, tabular and graphical summaries of all data collected during the year;
2. A discussion of compliance and the corrective actions taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements;
3. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program;

4. A copy of the certification for each certified wastewater treatment plant operator working at the facility and a statement about whether the Discharger is in compliance with Title 23, CCR, Division 3, Chapter 26;
5. Summary of information on the disposal of sludge and/or solid waste;
6. The results from any sludge monitoring required by the disposal facility; and
7. The results from annual monitoring of the effluent, groundwater and water supply.

A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the Discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain the penalty of perjury statement by the Discharger, or the Discharger's authorized agent, as described in the Standard Provisions General Reporting Requirements Section B.3.

The first monthly monitoring report required under MRP WQ 2014-0153-DWQ-R5279 is due on **1 December 2018** and is to cover the month of **October 2018**.

Ordered by:


For Patrick Pulupa, Executive Officer

22 August 2018

(Date)

Electronic Report Transmittal Form

Attention: Kenny Croyle (916) 464-4676
Discharger: East Bay Municipal Utility District
Pardee Reservoir Recreation Area Wastewater Treatment Plant
WDRs Order Number: WQ 2014-0153-DWQ-R5287
CIWQS Place ID: 247616
County: Amador

I am hereby submitting to the Central Valley Water Board the following information:

Check all that apply:

Technical Report Title and Date _____

Monthly Monitoring Report for the month of _____

1st/2nd (circle one) Semi-annual Monitoring Report for the year _____

Annual Monitoring Report for the year _____

Violation Notification:

During the monitoring period, there were / were not (circle one) any violations of the WDRs.

1. The violations were:

2. Have the violations been corrected? Yes / No. If no, what will be done to correct the violations:

Certification Statement:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

Signature: _____ Phone: _____

Printed Name: _____ Date: _____



**STATE WATER RESOURCES CONTROL BOARD
ORDER WQ 2014-0153-DWQ**

**GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS**

September 23, 2014



Water Boards

STATE WATER RESOURCES CONTROL BOARD
REGIONAL WATER QUALITY CONTROL BOARDS

September 23, 2014

STATE WATER RESOURCES CONTROL BOARD
ORDER WQ 20140-0153-DWQ
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

Contents

ACRONYMS AND ABBREVIATIONS	i
BACKGROUND INFORMATION	1
WASTEWATER DISPERSAL	5
APPLICATION PROCESS	8
ANTIDEGRADATION ANALYSIS	8
TITLE 27 EXEMPTION	11
CALIFORNIA ENVIRONMENTAL QUALITY ACT	12
OTHER REGULATORY ISSUES	13
IT IS HEREBY ORDERED	15
A. Prohibitions	15
B. Requirements by Wastewater System Type	16
1. All Wastewater Systems	16
2. Septic Systems	20
3. Aerobic Treatment Units	21
4. Activated Sludge Systems	23
5. Pond Systems	23
6. Subsurface Disposal Systems	24
7. Land Application and/or Recycled Water Systems	26
8. Sludge/Solids/Biosolids Disposal	26
C. Groundwater and Surface Water Limitations:	27
D. Effluent Limitations:	27
E. Provisions	28
1. Technical Report Preparation Requirements	28
2. For All Wastewater Systems:	30
3. General Reporting Requirements:	34
4. Monitoring Requirements	35
5. Definitions	37
CERTIFICATION	38

STATE WATER RESOURCES CONTROL BOARD
ORDER WQ 20140-0153-DWQ
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

ACRONYMS AND ABBREVIATIONS	
afy	acre feet per year
AGR	Agricultural Supply
Antidegradation Policy	State Water Board Resolution 68-16
APMP	Advanced Protection Management Program
AQUA	Aquaculture
Basin Plan	Water Quality Control Plan
BOD	Biochemical Oxygen Demand
BPTC	Best Practicable Treatment or Control
CalOES	California Office of Emergency Services
CDPH	California Department of Public Health
CEC	Constituents of Emerging Concern
CEQA	California Environmental Quality Act
C.F.R.	Code of Federal Regulations
DDW	State Water Board, Division of Drinking Water
E. coli	Escherichia coli
e.g.	Latin <i>exempli gratia</i> (for example)
FDS	Fixed Dissolved Solids
FEMA	Federal Emergency Management Agency
FRESH	Fresh Water Replenishment
General Order	General Waste Discharge Requirements Order
gpd	gallons per day
GWR	Groundwater Recharge
I/I	Inflow and Infiltration
IND	Industrial Service Supply
LAA	Land Application Area
MBR	Membrane Biological Reactor
MCL	Maximum Contaminant Level
mg/L	Milligrams per liter
MPI	Minutes Per Inch
MPN	Most Probable Number
MRP	Monitoring and Reporting Program
MUN	Municipal Supply
N	Nitrogen
NOA	Notice of Applicability
NPDES	National Pollutant Discharge Elimination System
NTU	Nephelometric Turbidity Unit
OWTS Policy	Onsite Wastewater Treatment System Policy
P	Phosphorus
pdf	Portable Document Format
Perc Rate	Percolation Rate
PROC	Industrial Process Supply
REC-1	Water Contact Recreation
Regional Water Board	Regional Water Quality Control Board

STATE WATER RESOURCES CONTROL BOARD
ORDER WQ 2014-0153-DWQ
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

BACKGROUND INFORMATION

Findings:

The State Water Resources Control Board (State Water Board) finds that:

1. Water Code section 13260(a) requires that any person discharging waste or proposing to discharge waste within any region, other than to a community sewer system, that could affect the quality of the waters of the state, file a Report of Waste Discharge (ROWD) to obtain coverage under Waste Discharge Requirements (WDRs) or a waiver of WDRs. "Waste" is defined in Water Code section 13050(d).
2. Discharges to land from Small Domestic Wastewater Treatment Systems (hereafter Small Domestic Systems) have certain common characteristics, such as similar constituents, concentrations of constituents, disposal techniques, flow ranges, and they require the same or similar treatment standards. These types of discharges are appropriately regulated under a General Waste Discharge Requirements Order (General Order). State Water Board Water Quality Order 97-10-DWQ (WQO 97-10-DWQ) is a 1997 General Order addressing Small Domestic Systems. Once effective, this General Order will supersede WQO 97-10-DWQ which will no longer be available for additional enrollees.
3. For the purposes of this General Order, the term "wastewater system" shall mean the collection system, treatment equipment, pumping stations, treatment ponds, clarifiers, sand/media filters, disinfection systems, recycled water systems (including distribution systems), storage ponds, land application areas, and other systems associated with the collection, treatment, storage, and disposal of wastewater.
4. Only Small Domestic Systems, with a monthly average flow rate of 100,000 gallons per day (gpd) or less, that discharge to land are eligible for coverage under this General Order. Small Domestic Systems are typically located at individual residences, rural parks, schools, campgrounds, mobile home parks, roadside rest stops, small commercial or residential subdivisions, restaurants, resort hotels/lodges, small correctional facilities, temporary fire-fighting camps, and recreational vehicle (RV) dump locations, including RV parks. An owner and/or operator of such a wastewater system is hereafter referred to as Discharger in this General Order. A Small Domestic System that uses subsurface disposal may be regulated by a local agency rather than a Regional Water Board, consistent with the Water Quality Control Policy for Siting, Design, Operation, and Maintenance of Onsite Wastewater Treatment Systems Policy (OWTS Policy). Wastewater systems regulated by local agencies may continue that coverage unless directed by the local agency or the Regional Water Board Executive Officer to seek WDRs from the Regional Water Board.
5. Wastewater treatment technologies evolve over time. Septic tanks and gravity fed leach fields provide the lowest level of acceptable treatment. Other treatment may include aerobic treatment systems, sand/media filters, package treatment plants, constructed wetlands, activated sludge, membrane biological reactors, and disinfection systems. Similarly, other dispersal options for the treated effluent may include pressure dosing, drip irrigation, land application, mound/at grade systems, or

STATE WATER RESOURCES CONTROL BOARD
ORDER WQ 2014-0153-DWQ
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

- 3 -

<u>Constituent</u>	<u>Units^a</u>	Typical Domestic Wastewater	Septic Tank Influent	Septic Tank Effluent	Secondary Treatment Effluent	Equivalent to Secondary Treatment Effluent
Nitrite and Nitrate (as N)	mg/L	<1 ^b	<1 ^c	-- ^{g,o}	-- ^{g,h}	-- ^{g,h,i}
Total Phosphorus (as P)	mg/L	6-12 ^b	6-12 ^c	5-15 ^d	51% ^m	50% ^{k,h,i}

^a. mg/L denotes milligrams per liter.

^b. Data from Table 4-3, USEPA Wastewater Treatment/Disposal for Small Communities, Manual, September 1992, EPA/625/R-92/005.

^c. Data from Table 3-7, USEPA Onsite Wastewater Treatment System Manual, June 2005, EPA/625/R-00/008.

^d. Data from Table 3-19, USEPA Onsite Wastewater Treatment Systems Manual, June 2005, EPA/625/R-00/008.

^e. Data from Exhibit 5-6, USEPA NPDES Permit Writers' Manual, December 1996, EPA-833-B-96-003.

^f. Data from Section 5.2.2, USEPA NPDES Permit Writers' Manual, December 1996, EPA-833-B-96-003.

^g. "--" denotes data not available.

^h. Value highly variable depending upon treatment technology.

ⁱ. No technology based limit established by USEPA.

^k. Percent reduction from influent, wastewater pond treatment. USEPA webpage <http://www.epa.gov/caddis/ssr_urb_ww1.html>, accessed August 29, 2014.

^m. Percent reduction from influent wastewater, activated sludge treatment. USEPA webpage <http://www.epa.gov/caddis/ssr_urb_ww1.html>, accessed August 29, 2014.

^o. Insignificant change expected in treatment.

^p. TSS limit not appropriate for land discharge.

9. Discharges from RV holding tanks or portable toilets may contain chemicals that can pollute groundwater quality. Some commercially available products used to control holding tank/portable toilet odors may contain harmful chemicals such as formaldehyde, zinc, or phenol. The harmful chemicals can kill the bacteria in the wastewater treatment system and cause wastewater to be inadequately treated. Inadequately treated wastewater may cause additional problems such as leachfield/seepage pit failure, surfacing wastewater, and potential exposure and health risks. Discharge of the harmful chemicals to groundwater that creates pollution may result in enforcement activities requiring groundwater remediation. The best and least expensive method to prevent groundwater pollution is to not use harmful chemicals by educating RV owners about the pollution hazard.

10. Total dissolved solids (TDS) consists of both volatile (organic) and fixed (inorganic) fractions. A varying concentration of volatile dissolved solids will exist in wastewater effluent depending upon the wastewater source and treatment technology. In a properly operated land application system, volatile dissolved solids in percolate are generally reduced to negligible concentrations (less than 2 mg/L) by filtration and biological degradation following percolation through five feet of soil.¹ However, fixed

¹ USEPA, Process Design Manual, Land Treatment of Municipal Wastewater, Section 4.2.1, 1981.

14. The Recycled Water Policy includes monitoring requirements for Constituents of Emerging Concern³ for the use of recycled water for groundwater recharge by surface and subsurface application methods. The monitoring requirements and criteria for evaluating monitoring results in the Recycled Water Policy are based on recommendations from a Science Advisory Panel.⁴ Because this General Order is limited to non-potable uses and does not authorize groundwater replenishment activities, monitoring for Constituents of Emerging Concern is not required.
15. The Recycled Water Policy requires permits for landscape irrigation with recycled water to include priority pollutant monitoring at the recycled water production facility. Annual monitoring is required for design production flows greater than one million gallons per day; a five year monitoring frequency is required for flows less than one million gallons per day. Priority pollutants are listed in Appendix A of 40 Code of Federal Regulations (C.F.R.) Part 423.
16. Beneficial uses for groundwater are determined by each Regional Water Board and are listed in their respective Basin Plans. Beneficial uses for groundwater are: municipal supply (MUN), industrial service supply (IND), industrial process supply (PROC), fresh water replenishment (FRESH), aquaculture (AQUA), wildlife habitat (WILD), water contact recreation (REC-1), agricultural supply (AGR), and groundwater recharge (GWR). Some beneficial uses only apply to certain geographic areas within regions.
17. Basin Plans establish groundwater quality objectives to protect beneficial uses. The objectives may be narrative, numerical, or both. This General Order requires the Discharger to comply with those objectives in receiving groundwater.

WASTEWATER DISPERSAL

18. Wastewater dispersal will occur by different methods. It may be percolated from ponds; applied to the surface by spray, flood, or drip methods; or discharged to a subsurface dispersal area such as a leachfield or seepage pit. The choice of disposal method will depend upon the amount of wastewater generated, the value of the wastewater for irrigation, and the receiving environment.
 - a. Wastewater discharged to a pond for treatment and/or storage can result in groundwater degradation or nuisance odors. Ponds can also be vulnerable to damage caused by burrowing animals.

³ For this order, Constituents of Emerging Concern are defined to be chemicals in personal care products, pharmaceuticals including antibiotics, antimicrobials; industrial, agricultural, and household chemicals; hormones; food additives; transformation products, inorganic constituents; and nanomaterials.

⁴ The Science Advisory Panel was convened in accordance with provision 10.b of the Recycled Water Policy. The panel's recommendations were presented in the report; *Monitoring Strategies for Chemicals of Emerging Concern in Recycled Water - Recommendations of a Science Advisory Panel*, dated June 25, 2010.

system, and should be constructed with cleanouts or a flushing system similar to at-grade and/or above grade systems when needed.

Subsurface disposal areas should be planted with shallow rooted plants to prevent erosion and provide for uptake of wastewater nutrients; trees and shrubs should be removed to prevent roots from damaging the leachfield. Similarly, burrowing animals can damage an at-grade or above grade (mound) disposal system and result in leakage. Burrowing animals should be promptly controlled and repairs to the disposal system completed as soon as possible.

19. Setbacks from wastewater treatment areas, dispersal areas, and/or LAAs from domestic wells, flowing and/or ephemeral streams, lakes/reservoirs, and property lines are provided in this General Order. Setbacks are included as a means of reducing pathogenic risks by coupling pathogen inactivation rates with groundwater travel time to a well or other potential exposure route (e.g. water contact activities). In general, a substantial unsaturated zone reduces pathogen survival compared to saturated soil conditions. Fine grained (silt or clay) soil particles reduce the rate of groundwater transport and therefore are generally less likely to transport pathogens; coarse grained soil particles or fracture flow groundwater conditions may be more likely to transport pathogens. Setbacks also provide attenuation of other wastewater constituents through physical, chemical, and biological processes. The setbacks provided in this General Order are based on the title 22 water recycling criteria, the California Well Standards, the OWTS Policy, the California Plumbing Code, and commonly imposed setbacks by regulatory agencies.
20. The OWTS Policy identified wastewater disposal as a potential contributing source of pathogens or nitrogen to an impaired surface water body.⁵ OWTS Policy Tier 3 addresses impaired surface water bodies and describes an Advanced Protection Management Program (APMP) as the minimum management program for wastewater systems subject to the OWTS Policy. In general, wastewater systems located within the geographic area⁶ of an APMP are not eligible for coverage under this General Order unless the Regional Water Board's Executive Officer determines the discharge is acceptable based on site-specific conditions, the level of wastewater treatment, and/or total maximum daily load (TMDL) implementation plan requirements.

⁵ Impaired surface water bodies are those identified on a list approved first by the State Water Board and then approved by USEPA pursuant to Section 303(d) of the Federal Clean Water Act.

⁶ The geographic area of an APMP is established (in order of hierarchy) by an approved TMDL, an approved local agency defined APMP, or 600 feet from the water body.

- a. Higher quality water will be maintained until it has been demonstrated to the state that any change will be consistent with the maximum benefit to the people of the state, will not unreasonably affect present and anticipated beneficial use of the water, and will not result in water quality less than that prescribed in the policies.
 - b. Any activity that produces a waste and discharges to existing high quality waters will be required to meet WDRs that will result in the Best Practicable Treatment or Control (BPTC) of the discharge necessary to assure pollution or nuisance will not occur, and the highest water quality consistent with the maximum benefit to the people of the state will be maintained.
25. The Antidegradation Policy requires maintenance of high quality of waters of the state unless limited degradation is consistent with the maximum benefit to the people of the state. When issuing NOAs under this General Order, Executive Officers must assure that Dischargers implement BPTC as necessary to maintain the highest water quality consistent with the maximum benefit to the people of the state.
26. This General Order allows discharges to numerous groundwater bodies, each with its own chemical characteristics. There are not sufficient data to determine which receiving waters are high quality waters. To the extent a discharge covered under this General Order may be to high quality waters, this General Order authorizes limited degradation consistent with the Antidegradation Policy as described in the findings below.
27. This General Order limits a discharge flow rate to 100,000 gpd; therefore, only small Dischargers will be eligible for coverage. Wastewater systems with a flow rate greater than 20,000 gpd must evaluate the discharge with the method presented in Attachment 1 to determine if nitrogen effluent limits are appropriate. Discharge of domestic wastewater at lower flow rates inherently has less potential to significantly degrade water quality.
28. Limited degradation of groundwater by some waste constituents associated with domestic wastewater effluent, after effective source control, treatment, and control measures are implemented, is consistent with the maximum benefit to the people of the state. The technology, energy, water recycling, and waste management advantages of centralized wastewater treatment systems far exceed any benefits derived from reliance on numerous, concentrated individual wastewater systems, and the cumulative impact on water quality will be substantially less. The economic prosperity of communities and associated industry is of maximum benefit to the people of the state and provides sufficient justification for allowing the limited groundwater degradation that may occur pursuant to this General Order provided the terms of the applicable Basin Plan, and other applicable State Water Board and Regional Water Board policies are consistently met. The State Water Board recognizes variability in a small community's ability to construct wastewater treatment systems based on financial resources.
29. Constituents of concern that have the potential to degrade groundwater include salinity, nutrients, and pathogens (represented by coliform bacteria). In addition, excessive BOD loading of ponds or LAAs may result in nuisance odors or anaerobic

pathogenic bacteria, and are easy to detect and quantify. Disinfection of wastewater is not required in every situation, such as when the wastewater application is performed in such a way that public contact is minimized through physical controls and/or notification.

When needed, disinfection can be performed in a number of ways. The title 22 water recycling criteria lists disinfection requirements for specific activities. To ensure the disinfection is effective, the model MRP provided as Information Sheet Attachment C includes monitoring that can be implemented to verify compliance with effluent limits.

30. Compliance with the General Order, the NOA, DDW requirements, and any mitigation measures will ensure compliance with the applicable Basin Plan.

TITLE 27 EXEMPTION

31. The wastewater treatment, storage, and disposal activities described in this General Order are exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste* in California Code of Regulations, title 27, division 2, Subdivision 1, section 20005, et seq. The activities are exempt from the requirements of title 27 so long as the activity meets, and continues to meet, all preconditions listed below. (Cal. Code Regs., tit. 27, § 20090.)
- a. Sewage—Discharges of domestic sewage or treated effluent which are regulated by WDRs issued pursuant to California Code of Regulations, title 23, division 3, chapter 9, or for which WDRs have been waived, and which are consistent with applicable water quality objectives, and treatment or storage facilities associated with municipal wastewater treatment plants, provided that residual sludge or solid waste from wastewater treatment facilities shall be discharged only in accordance with the applicable State Water Board promulgated provisions of this division. (Cal. Code Regs., tit. 27, § 20090(a).)
 - b. Wastewater—Discharges of wastewater to land, including but not limited to evaporation ponds, percolation ponds, or subsurface leach fields if the following conditions are met:
 - (1) the applicable Regional Water Board has issued WDRs, reclamation requirements, or waived such issuance;
 - (2) the discharge is in compliance with the applicable water quality control plan; and
 - (3) the wastewater does not need to be managed according to, California Code of Regulations, title 22, division 4.5, chapter 11, as a hazardous waste. (Cal. Code Regs., tit. 27, § 20090(b).)
 - c. Underground Injection—Discharges of waste to wells by injection pursuant to the Underground Injection Control Program established by the USEPA under the Safe Drinking Water Act, 42 US Code section 300(h), see Code of Federal Regulations title 40, Parts 144 to 146. (Cal. Code Regs., tit. 27, § 20090(c).)

- b. For new or expanding Small Domestic Systems, the State Water Board considered the environmental impacts associated with the adoption of this General Order and prepared an Initial Study in accordance with California Code of Regulations, title 14 section 15063. Analysis in the Initial Study and early consultation with responsible and trustee agencies did not identify any significant impacts on the environment. Therefore, a Negative Declaration was prepared. The State Water Board adopted the Negative Declaration (Resolution 2014-0054) on September 23, 2014.
- c. New or expanding systems are subject to further CEQA evaluation on a site-specific basis by local agencies performing CEQA evaluations of proposed projects. The potential significant environmental impacts from discharges of domestic wastewater can be mitigated to less than significant impacts by compliance with this General Order, the NOA, and any mitigation measures adopted by local agencies.

OTHER REGULATORY ISSUES

- 33. Dischargers that meet the criteria for coverage under State Water Board Order No. 2006-0003-DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems*, or updated order, are required to obtain coverage.
- 34. Consistent with Water Code section 13241, the State Water Board, in establishing the requirements contained herein, considered factors including, but not limited to, the following:
 - a. Past, present, and probable future beneficial uses of water.
 - b. Environmental characteristics of the hydrographic unit under consideration, including the quality of water available thereto.
 - c. Water quality conditions that could reasonably be achieved through the coordinated control of all factors which affect water quality in the area.
 - d. Economic considerations.
 - e. The need for developing housing within the region(s).
 - f. The need to develop and use recycled water.
- 35. Water Code section 13263(i) states, The State Water Board or a Regional Water Board may prescribe general WDRs for a category of discharges if the State Water Board or that Regional Water Board finds or determines that all of the following criteria apply to the discharges in that category:
 - 1) The discharges are produced by the same or similar operations.
 - 2) The discharges involve the same or similar types of waste.
 - 3) The discharges require the same or similar treatment standards.
 - 4) The discharges are more appropriately regulated under general WDRs than individual WDRs.

38. This General Order does not preempt or supersede the authority of municipalities, flood control agencies, or other local agencies to prohibit, restrict, or control discharges of waste subject to their jurisdiction.
39. The State Water Board has notified potential Dischargers and all other known interested parties of the intent to prescribe WDRs as described in this General Order.
40. The State Water Board, in a public meeting, has heard and considered all comments pertaining to the proposed discharge.

IT IS HEREBY ORDERED

IT IS HEREBY ORDERED that upon adoption of this General Order, WQO 97-10-DWQ is classified as in effect, but inactive for future Discharger enrollment.

Pursuant to Water Code section 13263 and 13267, the Discharger, its agents, successors, and assigns, in order to meet the provisions contained in division 7 of the Water Code and regulations adopted hereunder, shall comply with the following:

A. Prohibitions

1. The direct or indirect discharge of any wastewater to surface waters or surface water drainage courses is prohibited.
2. The use of cesspools, an excavation or device that allows wastewater infiltration into the soil without treatment, is prohibited.
3. The treatment, storage, and/or disposal of waste in or at the wastewater system shall not cause or contribute to a condition of pollution, contamination, or nuisance as defined in Water Code section 13050.
4. The discharge of wastewater other than domestic wastewater is prohibited.
5. Bypass or overflow of treated or untreated waste is prohibited.
6. The discharge of waste to land not owned, operated, or controlled by the Discharger is prohibited. An exception to this prohibition is when recycled water is used as described in a title 22 Engineering Report approved by DDW.
7. The discharge of waste classified as hazardous (Cal. Code Regs., tit. 23, § 2521(a)), or designated (Wat. Code, § 13173) is prohibited.
8. The discharge of waste in violation of, or not consistent with, the applicable Regional Water Board's Basin Plan is prohibited.
9. A physical connection between a recycled water system and a potable water system is prohibited.
10. The use of recycled water in a manner different than described in the DDW approved title 22 engineering report is prohibited.
11. Use of equipment used to convey recycled water (e.g. tanks, piping, valves,) also used for potable water supply, is prohibited.

- d. The siting, design, construction, operation, maintenance, and monitoring of the wastewater system shall comply with the requirements of the applicable Regional Water Board's Basin Plan.
- e. Nuisance odors shall not be perceivable beyond the property line of the wastewater treatment facility.
- f. Public contact with wastewater shall be deterred through such means as fences, signs, and other acceptable alternatives.
- g. For systems with a design flow rate greater than 3,500 gpd, the technical report required as part of the ROWD shall be prepared by a California licensed professional civil engineer. For systems with a design flow rate less than 3,500 gpd, the technical report shall be prepared by a California licensed professional engineer or other appropriately licensed professional (e.g., a California licensed professional geologist or California registered environmental health specialist).
- h. For new or expanding Small Domestic Systems within or nearby the boundaries of a centralized wastewater district or regional service area, the Discharger must demonstrate a good faith effort to connect to the centralized system when feasible and practicable, and provide evidence that connection to the system was not approved.
- i. A Regional Water Board Executive Officer may require additional investigations or monitoring to demonstrate beneficial uses of water are protected and antidegradation requirements are satisfied. Acceptable methods may include, but not be limited to, evaluation of the wastewater system's treatment performance, groundwater monitoring, or additional sampling to characterize the wastewater discharge.
- j. The Discharger shall comply with any water quality related mitigation measures adopted in a CEQA document addressing the facility.
- k. When producing or using recycled water, the Discharger shall comply with the provisions of the DDW approved title 22 Engineering Report.
- l. The Discharger shall comply with the setbacks described in Table 3. However, some existing sites may not comply with the setbacks provided herein. Such noncomplying sites may be permitted under this General Order if nuisance conditions do not result from the noncompliance. Expansion of a noncomplying wastewater system shall trigger further evaluation of the setbacks. In some cases, more than one setback standard exists. The following procedure shall be implemented when determining the appropriate setback:
 - i. When the setback requirement comes from title 22, approval of a variance must be obtained from DDW.

Table 3: Summary of Wastewater System Setbacks

Equipment or Activity	Domestic Well	Flowing Stream ^a	Ephemeral Stream Drainage ^b	Property Line	Lake or Reservoir ^d
Septic Tank, Aerobic Treatment Unit, Treatment System, or Collection System ^e	150 ft. ^y 100 ft. ^o 50 ft. ^c	50 ft. ^c	50 ft.	5 ft. ^c	200 ft. ^w 50 ft. ^c
Leach Field ^f	100 ft. ^{o,c}	100 ft. ^c	50 ft.	5 ft. ^c	200 ft. ^w 100 ft. ^c
Seepage Pit	150 ft. ^{o,c}	150 ft. ^c	50 ft.	8 ft. ^c	200 ft. ^w 150 ft. ^c
LAND APPLICATION AREA REQUIREMENTS					
LAA (disinfected tertiary recycled water) ^g	50 ft. ^m	25 ft.	50 ft.	25 ft.	200 ft.
LAA (disinfected sec-2.2 or sec-23 recycled water) ^h	100 ft. ^r	50 ft.	50 ft.	100 ft. ^x 50 ft. ^p	200 ft.
LAA (undisinfected secondary recycled water) ⁱ	150 ft. ^s	100 ft.	100 ft.	100 ft. ^x 50 ft. ^p	200 ft.
Spray Irrigation (disinfected tertiary recycled water) ^k	No spray irrigation of any recycled water, other than disinfected tertiary recycled water, shall take place within 100 feet of a residence or a place where public exposure could be similar to that of a park, playground, or school yard.				
WASTEWATER STORAGE AND/OR TREATMENT PONDS					
Impoundment (disinfected tertiary recycled water) ^g	100 ft. ^t	100 ft.	100 ft.	50 ft.	200 ft.
Impoundment (disinfected sec-2.2 or sec-23 recycled water) ^h	100 ft. ^r	100 ft.	100 ft.	50 ft.	200 ft.
Impoundment (undisinfected secondary recycled water) ⁱ	150 ft. ^s	150 ft.	150 ft.	50 ft.	200 ft.

LAA denotes Land Application Area. Sec denotes secondary.

^a A flowing stream shall be measured from the ordinary high water mark established by fluctuations of water elevation and indicated by characteristics such as shelving, changes in soil character, vegetation type, presence of litter or debris, or other appropriate means.

^b Ephemeral Stream Drainage denotes a surface water drainage feature that flows only after rain or snow-melt and does not have sufficient groundwater seepage (baseflow) to maintain a condition of flowing surface water. The drainage shall be measured from a line that defines the limit of the ordinary high water mark (described in "a" above). Irrigation canals are not considered ephemeral streams drainage

with holding tank additives that may contain, among other chemicals, formaldehyde, zinc, and/or phenol.

Use of holding tank chemicals shall be discouraged by the wastewater system owner/operator. Education of visitors can be accomplished by providing an information sheet upon check-in. Information on holding tank chemicals is provided in Attachment B2. The sale or provision of such additives by the Discharger to operators of RVs served by the Discharger's wastewater facility may be determined to be evidence of noncompliance with this section.

- c. To the maximum extent possible, RV, portable toilet, or similar wastes shall not be discharged to a septic tank or functionally equivalent system (e.g. Imhoff tank) without subsequent additional treatment (e.g., aerated pond, recirculating sand filter, etc.) prior to disposal.
- d. Septic tanks shall be pumped when any one of the following conditions exists:
 - i. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment.
 - ii. The scum layer is within 3 inches of the outlet device.
 - iii. The sludge layer is within 8 inches of the outlet device.
- e. Septage is the liquid, solid, and semisolid material that results from wastewater treatment in a septic tank, which must be pumped, hauled, treated, and disposed of properly. (40 C.F.R. § 503.) Septage disposal shall only be to a legal disposal site that has been issued WDRs by a Regional Water Board allowing septage disposal. Septage shall be handled in such a manner as to prevent its reaching surface waters or watercourses.

3. Aerobic Treatment Units

- a. Within **90 days** of the issuance of an NOA, the Discharger shall submit a Sludge Management Plan consistent with the requirements of Provision E.1.c of this General Order. The Discharger must obtain written approval from the Regional Water Board's Executive Officer prior to any disposal of sludge. The Executive Officer shall be notified of any changes in an approved Sludge Management Plan at least **90 days** in advance of the proposed change.
- b. Modifications to a Sludge Management Plan deemed part of an emergency action shall be noticed to the Regional Water Board Executive Officer within **five days** of disposal with a rationale for the emergency modification.
- c. Gauging and limited repairs may be performed by homeowners or contractors as allowed by the Business and Professions Code (Bus. & Prof. Code, §§ 7044, 7048). With certain exceptions, anyone performing construction work in California must be licensed by the California Contractors' State License Board. Septic tank, aerobic treatment unit, and/or leach field service (repairs, pumping, etc.) shall be performed only by a California licensed General Engineering (A), Plumbing (C-36), or Sanitation System (C-42) contractor. The Discharger shall maintain a record of all service activities for a minimum of five years. At a minimum, the record shall

of properly. (40 C.F.R. § 503.) Septage disposal shall only be to a legal disposal site that has been issued WDRs by a Regional Water Board allowing septage disposal. Septage shall be handled in such a manner as to prevent its reaching surface waters or watercourses.

4. Activated Sludge Systems

- a. Within **90 days** of the issuance of an NOA, the Discharger shall submit a Sludge Management Plan consistent with the requirements of Provision E.1.c of this General Order. The Discharger must obtain written approval from the Regional Water Board's Executive Officer prior to any disposal of sludge. The Executive Officer shall be notified of any changes in an approved Sludge Management Plan at least **90 days** in advance of the proposed change.
- b. Modifications to a Sludge Management Plan deemed part of an emergency action shall be noticed to the Regional Water Board's Executive Officer within **five days** of disposal with a rationale for the emergency modification.

5. Pond Systems

- a. Sufficient freeboard shall be maintained at all times in ponds to provide adequate storage capacity and prevent wastewater spills. Freeboard shall be measured vertically from the lowest elevation of the pond berm to the pond water surface. If freeboard is less than one foot, the discharger shall immediately implement the contingency plan contained in the *Spill Prevention and Emergency Response Plan* (Provision E.1.a).
- b. Pond systems shall have sufficient capacity to accommodate wastewater, design seasonal precipitation, ancillary I/I, and wind driven waves. Design seasonal precipitation shall be based on the following precipitation criteria:
 - i. If wastewater spills do not occur, existing pond systems may continue to operate at their present size if they are covered under individual WDRs, a general order issued by a Regional Water Board, or by WQO 97-10-DWQ. If wastewater spills do occur, the Executive Officer may require the pond size requirement to be consistent with the specification defined in Pond Systems 5.b.ii (below).
 - ii. For new or expanding pond systems covered under this General Order, seasonal precipitation used in the pond sizing water balance calculations shall be based on the following:
 - a. The 100-year return annual total precipitation value distributed monthly in accordance with average (mean) precipitation values. The calculations shall demonstrate adequate capacity to maintain two feet of freeboard in the pond(s).
 - b. The Executive Officer may allow a lower standard for the return annual total precipitation value, with approval of a technical report describing how operation of the wastewater system will not result in wastewater spills. In no case shall the Executive Officer allow less than a 50-year

- c. No part of the disposal system(s) shall extend to a depth where waste may pollute groundwater.
- d. All new or rehabilitated disposal areas associated with effluent pressure distribution systems (pressure-dosed systems) shall be equipped with cleanouts or a flushing system to allow solids to be removed from distribution pipes and orifices when needed.
- e. Deep rooted plants such as trees or shrubs shall be removed as needed from a subsurface disposal system area to prevent damage to the dispersal system by roots.
- f. Burrowing animals active in areas that may result in wastewater leakage from an at-grade or above grade (mound) disposal system shall be promptly controlled and repairs to the disposal system completed as soon as possible.
- g. Subsurface disposal systems including leach fields and seepage pits, must comply with USEPA Underground Injection Control requirements when classified as a Class V well. Subsurface disposal systems with at least one of the following characteristics are classified as Class V wells:
 - i. The system has the capacity to serve 20 or more persons per day.
 - ii. The system receives wastewater other than domestic wastewater such as that generated by manufacturing, chemical processing, industrial fluid disposal, automotive repair, or recycling.
 - iii. The system receives sewage containing biological agents (such as wastewater from recreational vehicles or portable toilets).

Disposal systems that are classified as Class V wells must be registered with USEPA either by completing the online form at: <http://www.epa.gov/region09/water/groundwater/injection-wells-register.html> , or by completing and submitting Form 7520-16: Inventory of Injection Wells. Form 7520-16 is available at: <http://epa.gov/region09/water/groundwater/uic-pdfs/7520-16.pdf> .

- h. Limited repairs may be performed by homeowners or contractors as allowed by the Business and Professions Code (Bus. & Prof. Code, §§ 7044, 7048). With certain exceptions, anyone performing construction work in California must be licensed by the California Contractors' State License Board. Leach field repairs shall be performed only by a California licensed General Engineering (A), Plumbing (C-36), or Sanitation System (C-42) contractor. The Discharger shall maintain a record of all repair activities for a minimum of five years. At a minimum, the record shall include the date, nature of repair, service company name, and service company state contractor license number.

sludge that will not be subject to further treatment at the wastewater system. Biosolids refers to sludge that has undergone sufficient treatment and testing to qualify for reuse pursuant to the USEPA Part 503 Biosolids Rule. (40 C.F.R. § 503.)

- b. Sludge and solid waste shall be removed from screens, sumps, tanks, and ponds as needed to ensure optimal plant operation.
- c. Treatment and storage of sludge shall be confined to the wastewater system property, and shall be conducted in a manner that precludes infiltration of waste constituents into soil in a mass or at concentrations that will violate the groundwater limitations of this General Order.
- d. Any storage of residual sludge, solid waste, or biosolids at the wastewater system shall be temporary, and the waste shall be controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or at concentrations that will violate the groundwater limitations of this General Order.
- e. Residual sludge, and solid waste shall be disposed of in a manner approved by the appropriate Regional Water Board's Executive Officer and consistent with the Consolidated Requirements for Treatment, Storage, Processing, or Disposal of Solid Waste. (Cal. Code Regs., tit. 27 div. 2.) Removal for further treatment, disposal, or reuse at disposal sites operated in accordance with valid WDRs issued by the State Water Board or Regional Water Board will satisfy this specification.
- f. Use and disposal of biosolids shall comply with the USEPA Part 503 Biosolids Rule. (40 C.F.R. § 503.)

C. Groundwater and Surface Water Limitations

1. The discharge shall not

- a. Pollute groundwater or surface waters.
- b. Adversely affect beneficial uses of groundwater or cause an exceedance of any applicable Basin Plan water quality objectives for groundwater or surface water.

D. Effluent Limitations

1. The discharge shall not:

- a. Exceed any of the applicable effluent limitations presented in Table 4. Effluent limitation selection is a two-step process. Step one is based upon the treatment technology employed; step two applies only to systems with a flow rate greater than 20,000 gpd and is based upon an evaluation of the need for a nitrogen effluent limit (as described in Attachment 1). Low and high threat are defined in Attachment 1. The limits presented below are average monthly limits unless otherwise specified.

- ii. Sludge Handling - A description of the sludge handling equipment, operational controls, and disposal procedures.
- iii. Collection System Maintenance - A description of collection system cleaning and maintenance, equipment tests, and alarm functionality tests to minimize the potential for wastewater spills originating in the collection system or headworks. For collection systems subject to State Water Board Order No. 2006-0003-DWQ, reports prepared to comply with the State Water Board Order No. 2006-0003-DWQ satisfy this requirement.
- iv. Emergency Response - A description of emergency response procedures including for emergencies such as power outage, severe weather, flooding, or inadequate freeboard (for systems with wastewater or recycled water ponds). An equipment and telephone list for contractors/consultants, emergency personnel, and equipment vendors.
- v. Notification Procedures - Coordination procedures with fire, police, Governor's Office of Emergency Services (CalOES), Regional Water Board, and local county health department personnel.

The Response Plan shall be maintained at the treatment facility and shall be presented to the Regional Water Board staff upon request.

- b. **Within 90 days** of the issuance of an NOA containing an MRP, the Discharger shall prepare and implement a written sampling and analysis plan (SAP) sufficient to assure compliance with the terms of this General Order and the NOA. Anyone performing sampling on behalf of the Discharger shall be familiar with the SAP. SAPs shall address the need for sample filtration and how filtration will be accomplished. When sampling groundwater or liquid waste, the chemical constituents available to migrate shall be considered. In general, dissolved waste constituents can migrate through soil to groundwater or surface water. In cases where the waste only threatens groundwater, samples shall be filtered prior to chemical preservation, digestion, or analysis for some analytes. If overland flow of liquid waste to surface water is possible, the total constituent concentrations may be available for movement and analyses shall be conducted on unfiltered samples.

At a minimum, the SAP shall describe the following:

- i. Sample chain-of-custody procedures and documentation.
- ii. Sampling locations.
- iii. Sampling frequencies.
- iv. Sample handling/preservation procedures.
- v. Analytical methods.
- vi. Sample containers, preservatives, and holding times.
- vii. For groundwater monitoring, well purging and field methods.

2. There were no feasible alternatives to bypass, such as the use of auxiliary treatment facilities or retention of untreated waste. This condition is not satisfied if adequate backup equipment or wastewater storage facilities should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that would otherwise occur during normal periods of equipment downtime or preventive maintenance; or
- ii. Scheduled Bypass
 1. Bypass is required for essential maintenance to assure efficient operation,
 2. Neither effluent nor groundwater limitations are exceeded,
 3. The Discharger notifies the Regional Water Board's Executive Officer **10 days** in advance, and
 4. The prohibition against discharge to surface water is not violated.
- b. A Discharger that wishes to establish the affirmative defense of an upset (see definition in Provision E.5.a) in an action brought for noncompliance shall demonstrate, through properly signed, contemporaneous operating logs, or other evidence, that all of the following is true:
 - i. An upset occurred and the cause(s) can be identified.
 - ii. The permitted facility was being properly operated at the time of the upset.
 - iii. The Discharger submitted notice of the upset as required in Provision E.3.a.
 - iv. The Discharger complied with any remedial measures required by this General Order, the NOA, or direction from the Regional Water Board's Executive Officer. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof.
- c. A Discharger whose wastewater flow rate has been increasing, or is projected to increase, shall estimate when the flow rate will reach hydraulic and treatment capacities of its treatment, collection, and disposal facilities. The projections shall be made in January, based on the last 3 years average dry weather flow rates, peak wet weather flow rates, and total annual flow rates, as appropriate. When any projection shows that capacity of any part of the facilities may be exceeded in 4 years, the Discharger shall notify the Regional Water Board's Executive Officer by March 1st. Providing the notification in an annual report is acceptable.
- d. The requirements prescribed herein do not authorize the commission of any act causing damage to the property of another, or protect the Discharger from liabilities under federal, state, or local laws. This General Order does not convey any property rights or exclusive privileges and does not create a vested right to continue to discharge wastewater.

- iv. A material change in the character, location, or volume of discharge.
- n. Before making a material change in the character, location, or volume of discharge, the Discharger shall notify the Regional Water Board Executive Officer. A material change includes, but is not limited to, any of the following:
 - i. An increase in area or depth used for waste disposal beyond that specified in the NOA.
 - ii. A significant change in disposal method, location, or volume (e.g., change from land application to percolation pond).

The Regional Water Board's Executive Officer may require that an ROWD be submitted.

- o. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or recycling areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this General Order, the Discharger shall notify the Regional Water Board's Executive Officer in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this General Order and the NOA.
- p. Except for material determined to be confidential in accordance with California law, all reports prepared in accordance with terms of this General Order shall be available for public inspection at the offices of the Regional Water Board. Data on waste discharges, water quality, geology, and hydrogeology are not confidential.
- q. The Discharger shall take all reasonable steps to minimize any adverse impact to waters of the state resulting from noncompliance with this General Order. Such steps shall include accelerated or additional monitoring as necessary to determine the nature and impact of the noncompliance.
- r. The Discharger shall maintain in good working order, and operate as efficiently as possible any facility, control system, or monitoring device installed to achieve compliance with this General Order and the NOA.
- s. The Discharger shall permit representatives of the Regional Water Board and/or the State Water Board, upon presentation of credentials, to:
 - i. Enter premises where wastes are treated, stored, or disposed of, and facilities in which any records are kept.
 - ii. Copy any records required under terms and conditions of this General Order.
 - iii. Inspect at reasonable hours, monitoring equipment required by this General Order.
 - iv. Sample, photograph, and/or video record any discharge, waste material, waste treatment system, or monitoring device.

- c. All reports submitted in response to this General Order, including monitoring reports, shall be signed by a person identified below:
- i. For a private residence: by the property owner of the residence.
 - ii. For a corporation: by a principal executive officer of at least the level of senior vice-president.
 - iii. For a partnership or sole proprietorship: by a general partner or the proprietor.
 - iv. For a municipality, state, federal, or other public agency: by either a principal executive officer or ranking elected or appointed official.
 - v. A duly authorized representative of a person described above if all of the following are completed:
 - 1) The authorization is made in writing by a person described above.
 - 2) The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity, such as the position of plant manager, operator of a waste management unit, superintendent, or position of equivalent responsibility. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.)
 - 3) The written authorization is submitted to the Regional Water Board.

Any person signing a document under this section shall make the following certification:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

- d. The Discharger shall mail a copy of each monitoring report and any other reports required by this General Order to the appropriate Regional Water Board or provide electronic submittals of reports or data as specified by the Regional Water Board. Contact and mail address information is available on the NOA or at:

http://www.waterboards.ca.gov/about_us/contact_us/docs/rwqcb directory.pdf

4. Monitoring Requirements

- a. The Discharger shall comply with the MRP issued with the NOA, and any future revisions, as specified by the appropriate Regional Water Board's Executive Officer. A model MRP is provided as Attachment C. However, the Executive Officer may modify or replace the MRP for site-specific treatment and disposal conditions when issuing the NOA, or revise the MRP when deemed necessary.

5. Definitions

- a. Upset means an exceptional incident in which there is unintentional and temporary noncompliance with effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper action.
- b. The monthly average flow rate is the total discharge by volume during a calendar month divided by the number of days in the month that the facility was discharging. This number shall be reported in gpd or million gallons per day.
- c. The monthly average concentration is the arithmetic mean of measurements recorded during a calendar month. If only one sample is collected in a calendar month, then that sample measurement is the monthly average concentration.
- d. The daily maximum concentration is the highest measurement recorded for any grab or composite sample collected during a day in a calendar month.
- e. The 7-day median total coliform organism value shall be calculated as the median concentration of the results for the last 7 calendar days. If only one sample is collected within a 7-day period, then that one sample becomes the 7-day median value.
- f. A grab sample is an individual sample collected in less than 15 minutes.
- g. Unless otherwise specified, a composite sample is a combination of individual samples collected over the specified sampling period. The method of compositing shall be reported with the results.
- h. A time-weighted sample is collected at equal time intervals, with a maximum interval of one hour.
- i. A flow-weighted sample is collected at varying time intervals (average interval one hour or less) so that each sample represents an equal portion of the cumulative flow. The duration of the sampling period shall be specified in the MRP.
- j. A day is the mean solar day of 24 hours beginning at mean midnight. All references to day in this General Order are calendar days.

ATTACHMENT 1
NITROGEN EFFLUENT LIMIT EVALUATION
ORDER WQ 2014-0153-DWQ
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

Introduction

The nitrogen effluent limit will only be imposed where required to protect beneficial uses of groundwater and surface water and shall not be selected as a default to add water quality protection where the added expense of nitrogen control is not required. It is the discharger's responsibility to provide adequate information to allow the evaluation of the need for additional treatment.

Each of the five site-specific considerations listed below shall be considered when evaluating a discharge and the need for nitrogen control. The site-specific conditions are further discussed on the explanation sheets. If each of the site-specific considerations is favorable (as discussed individually below), the facility is not required to meet an effluent limit for nitrogen.

The attached flow charts provide a method to evaluate the discharge and the receiving environment to determine the applicability of a nitrogen effluent limit. Evaluation of the need for a nitrogen effluent limit is a two-step process. In the first step, applicability of a nitrogen effluent limit is determined based on the flow rate and site-specific characteristics of the receiving environment; in the second step effluent limits are selected based on further evaluation of level of threat related to the site-specific characteristics of the discharge and the receiving environment.

To begin the evaluation start at "Step A: Flow and Site-Specific Considerations," on the following page.

Step A – Flow and Site-Specific Considerations

The following considerations shall be evaluated to determine if a nitrogen effluent limit is appropriate.

- A1 *Exceed 20,000 gpd Flow Rate?* The General Order allows a flow rate up to 100,000 gpd. However, discharges with flow rates less than 20,000 gpd are not required to meet a nitrogen effluent limit.
- A2 *Shallow Groundwater?* Shallow groundwater conditions are defined by the OWTS Policy Table 2, *Minimum Depth to Groundwater and Minimum Soil Depth from the Bottom of the Dispersal System* (reproduced as Table 5 below).

Table 5: Minimum Depth to Groundwater and Minimum Soil Depth from the Bottom of Dispersal System

<u>Percolation Rate</u> ^a	<u>Depth to Groundwater</u> ^b
Perc Rate ≤ 1 MPI	Additional Treatment Required
1 MPI ≤ Perc Rate ≤ 5 MPI	20 Feet
5 MPI ≤ Perc Rate ≤ 30 MPI	8 Feet
30 MPI ≤ Perc Rate ≤ 120 MPI	5 Feet
Perc Rate ≥ 120 MPI	Engineered Disposal Required

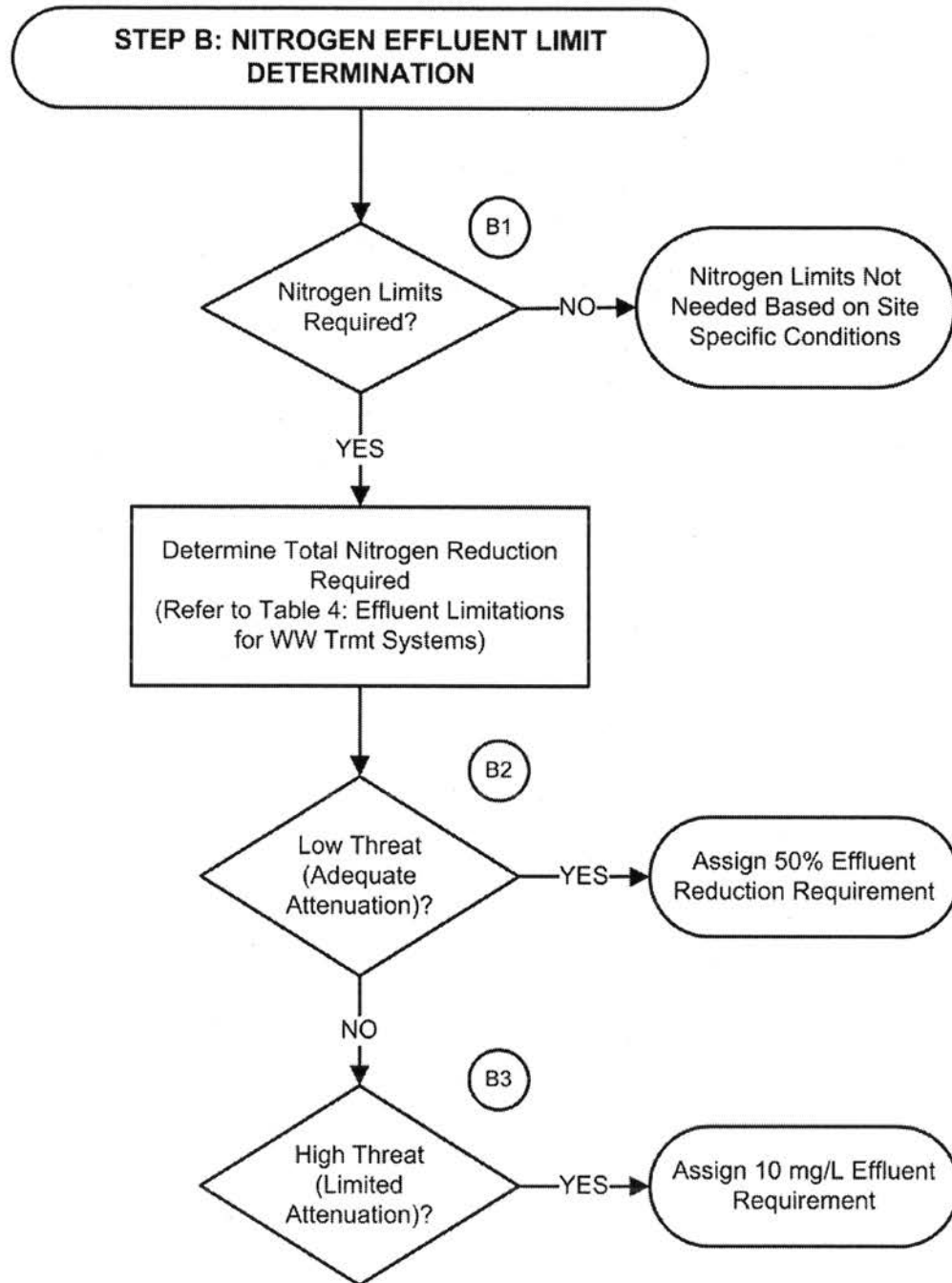
^a Perc Rate denotes percolation rate. MPI denotes minutes per inch.

^b The minimum depth to groundwater is measured from the base of the infiltration surface to the seasonally high groundwater table or first saturated interval.

- A3 *Excessive Percolation Rate / Fractured Environment?* An excessive percolation rate is defined as a combination of percolation rate and depth to groundwater that does not comply with the conditions presented in Table 5.

A fractured environment is defined as less permeable rock with porosity resulting from fractures that allows groundwater to flow through the fractures and has either of the following: a) No unconsolidated soil cover, or b) Unconsolidated soil cover that possesses an excessive percolation rate (that does not comply with the conditions defined in Table 5).

- A4 *Exceed Domestic Wastewater Strength?* Typical domestic wastewater strength is presented in Finding 8 of the General Order. Alternative domestic wastewater strengths may be based upon other per capita flow rate assumptions. The data presented in the General Order may be used; alternative characterizations from textbook and/or governmental organizations (e.g. U.S. Environmental Protection Agency) may be acceptable at the discretion of Regional Water Quality Control Board (Regional Water Board) staff.



INFORMATION SHEET SUMMARY
 STATE WATER RESOURCES CONTROL BOARD
 ORDER WQ 2014-0153-DWQ
 GENERAL WASTE DISCHARGE REQUIREMENTS
 FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

The attachments to this Information Sheet will help Dischargers understand the process of obtaining coverage under these *General Waste Discharge Requirements for Discharges to Land by Small Domestic Systems* (General Order). The attachments describe the permitting process and the information needed by Regional Water Quality Control Board (Regional Water Board) staff to prepare a Notice of Applicability (NOA), which provides the Discharger coverage under the General Order. Although not required, organization of the technical report described in Attachment B1 in the format presented in the attachment will allow streamlined review of the facility information and may reduce the time required to prepare an NOA. Some Regional Water Boards have implemented procedures for electronic submittal of technical reports and monitoring data. The Dischargers shall comply with those submittal requirements when applicable.

The Discharger is encouraged to contact the Regional Water Board staff early in the process to discuss their conceptual wastewater plan. Attachment A shows the generalized permitting process; it is recommended each of the items in the box labeled "Contact Regional Water Board to Discuss" be listed on a meeting agenda so that they are adequately discussed.

The *Conceptual Wastewater Plan* listed on Attachment A shall be complete enough for a meaningful discussion with the Regional Water Board staff so that any significant issues can be identified early in the process. However, the details of the *Conceptual Wastewater Plan* are unlikely to be finalized at this stage of the process. Determinations regarding the *Conceptual Wastewater Plan* may require additional investigation by the Discharger before the Regional Water Board staff can provide definitive answers to questions about the *Conceptual Wastewater Plan*.

Questions the Discharger may have regarding any of the attachments (such as the report requirements described in Attachment B1) should also be discussed at the meeting. At the conclusion of the meeting, the Discharger should understand how their system will be evaluated using the General Order Attachment 1, *Nitrogen Effluent Limit Evaluation*, and if any additional investigations are required to provide a complete Report of Waste Discharge.

The attachments included in this Information Sheet consist of the following:

<u>Name</u>	<u>Title</u>
Attachment A	Generalized Permit Application Process Summary
Attachment B1	Recommended Report of Waste Discharge Format
Attachment B2	Safe Wastewater Disposal for Recreational Vehicles
Attachment C	Model Monitoring and Reporting Program

ATTACHMENT B1 – INFORMATION SHEET
 RECOMMENDED REPORT OF WASTE DISCHARGE FORMAT
 STATE WATER RESOURCES CONTROL BOARD
 ORDER WQ 2014-0153-DWQ
 GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL DOMESTIC
 WASTEWATER TREATMENT SYSTEMS
 GENERAL INFORMATION FOR DISCHARGER

The information presented in the Report of Waste Discharge (ROWD) is relied upon by staff to prepare the Notice of Applicability (NOA) for coverage by this General Waste Discharge Requirements for Order (General Order). The Discharger shall ensure that the information presented in the ROWD is accurate. Misstatements, errors, or omissions that exist in the ROWD may be included in the NOA and become enforceable.

Waste Discharge Requirements (WDRs) are generally updated at 10 or 15 year intervals depending on the waste's potential to impact water quality. The ROWD shall state realistic growth projections. Underestimating growth may result in additional or more frequent permitting requirements. Overestimating growth will result in the need for the Discharger to prepare more treatment, storage, and disposal capacity than might otherwise be immediately required.

The ROWD outline presented below is intended to provide general guidance for Dischargers and consultants. Submitting an ROWD consistent with the format will help the Discharger include all of the information that Regional Water Quality Control Board (Regional Water Board) staff need and will expedite review of the document and speed the permitting process. Contacting your Regional Water Board representative to discuss the project before preparing the ROWD is recommended.

1. BACKGROUND

1.1. Wastewater system description

- 1.1.1. Briefly, describe what the wastewater system is and how wastewater is generated.
- 1.1.2. Provide a site location map and a site plan.
- 1.1.3. Provide information on the location of wastewater system buildings, wastewater treatment system components, groundwater wells, and surface water bodies.
- 1.1.4. Provide the Assessor's Parcel Number(s), section number(s), and Township and Range.
- 1.1.5. Describe the water supply to the residence(s), business(es), and/or other facilities being served by the wastewater system.

1.2. Service area description

- 1.2.1. Describe the proximity of the wastewater system to an existing regional collection system; if nearby, discuss why connection to the regional system cannot be accomplished. If located within a regional system service area, or in close proximity to a collection system, provide written documentation that a good faith effort to connect to the regional system was made and that the request was not approved.

- 2.2.2. Describe wastewater pretreatment components.
 - 2.2.2.1. Domestic wastewater pretreatment systems (e.g. septic tank effluent pump system, grease traps, etc.).
 - 2.2.2.2. Describe storage, treatment, and disposal of pretreatment residuals.
- 2.2.3. Describe preliminary treatment activities (e.g., screening, comminution, grit removal).
 - 2.2.3.1. Describe storage, treatment, and disposal of preliminary treatment residuals.
- 2.2.4. Describe primary treatment activities (remove settleable/flotable matter)
 - 2.2.4.1. Describe storage, treatment, and disposal of primary treatment residuals.
- 2.2.5. Describe treatment technology (e.g., activated sludge, membrane biological reactor, aerated lagoon, oxidation ditch, Imhoff tank, septic tank, etc.) include engineered design capacity in description.
 - 2.2.5.1. Describe storage, treatment, and disposal of treatment residuals (e.g. sludge, septage, etc.).
- 2.2.6. Size and location of treatment equipment (e.g. septic tank volume, package treatment plant, membrane biological reactor, pond size include acreage and storage capacity, pond liners, and number and horsepower of aerators, etc.).
- 2.2.7. Disinfection system equipment
- 2.2.8. Storage facilities
 - 2.2.8.1. If wastewater will be stored prior to disposal, describe the size and location of wastewater storage ponds, include a map showing all the ponds and describe them as lined or not. Describe the materials, age, and condition of any liners.
- 2.2.9. Predicted wastewater effluent quality
 - 2.2.9.1. Characterize the wastewater for TSS, BOD, total coliform organisms (if needed), and specific constituents of concern as needed. If RV waste is discharged to the system, characterize for holding tank chemicals identified in the General Order (and others as appropriate).
- 2.2.10. Treated effluent disposal method
 - 2.2.10.1. Describe how treated wastewater will be dispersed (land application area, leach field, percolation pond).
 - 2.2.10.2. Describe the proposed disposal area (and the 100-percent replacement area when needed, such as for a leach field disposal system) include acreage, surrounding land use, depth to groundwater, and the proximity of drainage ways, surface waters, and municipal, industrial, or agricultural wells.

- 2.3.1.1. Describe how any DDW requirements will be implemented in the project.
 - 2.3.1.2. If needed, describe the disinfection requirements for the planned reuse.
 - 2.4. Operation and Maintenance
 - 2.4.1. Describe routine operation and maintenance procedures
 - 2.4.2. Treatment operator training and qualifications requirements
 - 2.4.3. Contingency plans for repairs/spills/treatment issues
- 3. GROUNDWATER QUALITY
 - 3.1. Depending upon the threat to groundwater quality, groundwater monitoring may be required. Please contact your Regional Water Board representative to determine if groundwater monitoring is required.
- 4. CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)
 - 4.1. Some existing Small Domestic Systems will be determined to be categorically exempt from the California Environmental Quality Act (CEQA) under Title 14, section 15301 (ongoing or existing projects), section 15302 (replacement or reconstruction of existing utility systems), and section 15303 (new construction or conversion of small structures). The potential for categorical exemptions shall be examined and discussed with the Regional Water Board representative prior to submitting an ROWD.
 - 4.2. New or expanding Small Domestic Systems will likely require CEQA evaluation that should be performed by local agencies. The CEQA evaluation shall be submitted with the ROWD. At a minimum, the evaluation shall include the Initial Study, a list of any adopted mitigation measures related to water quality, and the Notice of Determination.
 - 4.2.1. The ROWD must include a description of how any water quality related mitigation measures will be implemented.
- 5. ADDITIONAL TECHNICAL REPORTS
 - 5.1. If required by the General Order, a *Sludge Management Plan* shall be submitted with the ROWD.
 - 5.1.1. Estimate the amount of sludge and scum that will be generated.
 - 5.1.1.1. Describe how sludge, scum, and supernatant will be stored and disposed of to protect groundwater quality.
 - 5.1.1.2. If sludge will be subject to further treatment, describe the treatment and storage requirements.
 - 5.1.1.3. Describe cleaning of digesters or storage vessels and the treatment and disposal of the residuals. If drying of residuals is planned, describe how that will be performed to prevent nuisance odors, prevent vectors, and protect groundwater quality.

**REPORT SEWAGE SPILLS
and other health hazards to the local
health department.
Keep People and Pets Away!**

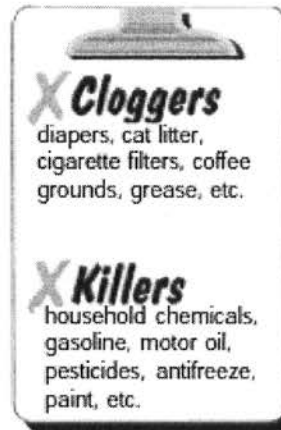
PARK OPERATORS:

The United States Department of Health, Education and Welfare said in 1957 that "... there are no known chemicals, yeasts, bacteria, enzymes or other substances capable of



eliminating or reducing the solids and scum in a septic tank" and according to EPA, this is still true. No products have been verified by EPA to eliminate the need for routine maintenance, and some may actually accelerate system failure by allowing solids to clog the dispersal system; while the products may claim to "remove" sludge, they may just "move" sludge. Tanks should be checked routinely (see photo) for solids and scum buildup.

Sludge Removal (pumpouts) may be needed more often for RV, Mobile Home and Boat waste systems than for single-family septic systems, especially if your tanks are undersized and/or your residents are conservative with water. Oversizing your tanks, or adding additional tanks, may allow greater waste stabilization. Consult a wastewater professional.



**RVers CAN HELP...
Here's How:**

- Minimize your need of holding tank deodorizers by using rest stop facilities when you can.

- If you must use a holding tank deodorizer, read the label carefully. Biodegradable (enzyme and citrus-based) products are available. Whichever product you

choose, follow label directions and add no more than recommended amounts.

- Some products that claim to be flushable, such as some types of cat litter, may clog hoses and septic tanks; use toilets for waste and toilet paper only.

- Ask questions of your park manager about drinking water and wastewater management. Sanitation costs can be minimal, but not free.

- Educate other RVers. Don't be shy about health.

FREE HOTLINES!

Septic System Care: The National Small Flows Clearinghouse, (800) 624-8301, EST, or www.nsfcc.org
The Safe Drinking Water Act Hotline, US EPA: (800) 426-4791, EST, or www.epa.gov/ogwdw

U.S. EPA, Region 9, WTR-9
Ground Water Office
75 Hawthorne Street
San Francisco, CA 94105-3109
OFFICIAL BUSINESS - PENALTY FOR PRIVATE USE \$300

The Discharger owns and operates the wastewater system that is subject to the Notice of Applicability (NOA) of Water Quality Order 2014-0153-DWQ. The reports are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to Water Code section 13267, the Discharger shall implement this MRP and shall submit the monitoring reports described herein.

All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample shall be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) shall be approved by Regional Water Board staff.

Field test instruments (such as those used to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Board California Environmental Laboratory Accreditation Program certified laboratory, or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

SEPTIC TANK MONITORING

Monitoring of septic tank shall include the following:

<u>Parameter</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow Rate	gpd	Metered ^a	Continuous	Annually

gpd denotes gallons per day.

^a. Flow rate may be metered or estimated based on potable water supply meter readings or other approved method.

Septic tanks shall be inspected and/or pumped at least as frequently as described below. Inspections of sludge and scum depth are not required if the tanks are pumped at least annually.

Effluent Monitoring

Samples of effluent shall be taken at an area that represents the effluent quality distributed to the disposal area. At a minimum, effluent monitoring shall consist of the following:

<u>Parameter</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow Rate	gpd	Metered ^a	Continuous	Quarterly
Biochemical Oxygen Demand	mg/L	Grab	Monthly	Quarterly
Total Nitrogen ^b	mg/L	Grab	Monthly	Quarterly

gpd denotes gallons per day. mg/L denotes milligrams per liter.

- a. Flow rate may be metered or estimated based on potable water supply meter readings or other approved method. Flow rates may be measured as influent or effluent flow.
- b. Include nitrogen monitoring when a nitrogen effluent limit is imposed.

Aerobic treatment units may be integrated in a treatment train and all components shall be inspected to verify operational status. It is highly recommended that a service agreement with a qualified service provider/vendor be required by the Regional Water Board's Executive Officer. Because aerobic treatment units generate more biosolids than septic systems (similar to the activated sludge process), systems shall be inspected and/or pumped at least as frequently as described below. Depending upon the amount of solids removed from the aerobic treatment unit, less frequent inspections may be allowed by the Regional Water Board's Executive Officer. Inspections of sludge and scum depth are not required if the tanks are pumped at least annually.

<u>Parameter</u>	<u>Units</u>	<u>Measurement Type</u>	<u>Inspection/Reporting Frequency</u>
Sludge depth and scum thickness in each compartment of each tank	Feet	Staff Gauge	Quarterly
Distance between bottom of scum layer and bottom of outlet device	Inches	Staff Gauge	Quarterly
Distance between top of sludge layer and bottom of outlet device	Inches	Staff Gauge	Quarterly
Effluent filter condition (if equipped, clean as needed)	NA	NA	Quarterly

NA denotes not applicable.

POND SYSTEM MONITORING ¹²

Influent Monitoring

Influent samples shall be taken from a location that provides representative samples of the wastewater and flow rate. At a minimum, influent monitoring shall consist of the following:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Flow Rate ^a	gpd	Meter	Continuous	Quarterly
Total Nitrogen ^b	mg/L	Grab	Monthly	Quarterly

gpd denotes gallons per day. mg/L denotes milligrams per liter.

- a. At a minimum, the total flow shall be measured monthly to calculate the average daily flow for the month. If wastewater is stored and applied to land, flow rate measurement may also be needed on the effluent flow.
- b. When needed for 50% reduction effluent limit calculation.

Wastewater Pond Monitoring

All wastewater and treated wastewater storage ponds (lined and unlined) shall be monitored as specified below:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Dissolved Oxygen	mg/L	Grab	Monthly	Quarterly
Freeboard	0.1 feet	Measurement	Monthly	Quarterly
Odors	--	Observation	Monthly	Quarterly
Berm condition	--	Observation	Monthly	Quarterly

mg/L denotes milligrams per liter.

Effluent Monitoring

Effluent samples shall be taken from a location that provides representative samples of the wastewater. At a minimum, effluent monitoring shall consist of the following:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Biochemical Oxygen Demand	mg/L	Grab	Monthly	Quarterly

¹² Determine the need for monitoring based on the flow rate and Attachment 1. Biochemical oxygen demand limits apply with flow rates above 400 gpd; nitrogen limits may apply at flow rates above 20,000 gpd. (See General Order Section D, Effluent Limits and Attachment 1, Nitrogen Effluent Limit Evaluation.)

<u>Constituent</u>	<u>Inspection Frequency</u>	<u>Reporting Frequency</u>
Pump Controllers, Automatic Valves, etc. ^a	Quarterly	Quarterly
Nuisance Odor Condition	Quarterly	Quarterly
Saturated Soil Conditions ^b	Quarterly	Quarterly
Plant Growth ^c	Quarterly	Quarterly
Vectors or Animal Burrowing ^d	Quarterly	Quarterly
Seepage Pit Condition ^e	Quarterly	Quarterly

- a. All pump controllers and automatic distribution valves shall be inspected for proper operation as recommended by the manufacturer.
- b. Inspect a disposal area for saturated conditions. If a mound system is used, inspect perimeter base for signs of wastewater seepage or saturated soil conditions.
- c. Shallow-rooted plants are generally desirable, deep-rooted plants such as trees shall be removed as necessary.
- d. Evidence of animals burrowing shall be immediately investigated and burrowing animal populations controlled as necessary.
- e. Seepage pits shall be inspected to ensure they are allowing wastewater to infiltrate as designed. Visual inspection of the water level in the seepage pit is adequate.

RECYCLED WATER MONITORING

If recycled water is used for irrigation of landscape areas, ¹³ priority pollutant monitoring is required at the production facility. Sampling shall be consistent with the following:

<u>Constituent</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Priority Pollutants	5 years	The next annual report.

mgd denotes million gallons per day.

LAND APPLICATION AREA MONITORING

The Discharger shall monitor LAAs when wastewater and/or supplemental irrigation water is applied. If wastewater/supplemental irrigation water is not applied during a reporting period, the monitoring report shall so state. LAA monitoring shall include the following:

¹³ Landscape areas are defined as parks; greenbelts, playgrounds; school yards; athletic fields; golf courses; cemeteries; residential landscaping; common areas; commercial landscaping (except eating areas); industrial landscaping (except eating areas); freeway, highway, and street landscaping

Prior to sampling, groundwater elevations shall be measured and the wells shall be purged of at least three well volumes and until pH and electrical conductivity have stabilized. No-purge, low-flow, or other sampling techniques are acceptable if they are described in an approved Sampling and Analysis Plan. Depth to groundwater shall be measured to the nearest 0.01 feet. Groundwater elevations shall be calculated. Samples shall be collected using approved USEPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling/Reporting Frequency</u> ^{c,d}
Groundwater Elevation ^a	0.01 Feet	Calculated	Quarterly
Depth to Groundwater	0.01 Feet	Measurement	Quarterly
Gradient	Feet/Feet	Calculated	Quarterly
Gradient Direction	degrees	Calculated	Quarterly
pH	Std. Units	Grab	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly
Nitrate as Nitrogen	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Total Coliform Organisms ^b	MPN/100 mL	Grab	Quarterly
Zinc ^c	mg/L	Grab	Quarterly
Phenol ^c	mg/L	Grab	Quarterly
Formaldehyde ^c	mg/L	Grab	Quarterly

MPN/100 mL denotes most probable number per 100 mL sample. Std. Units denotes standard units. mg/L denotes milligrams per liter.

- a. Groundwater elevation shall be based on depth to water using a surveyed measuring point elevation on the well and a surveyed reference elevation.
- b. Using a minimum of 15 tubes or three dilutions.
- c. Monitoring of the constituents zinc, phenol, and formaldehyde are required only when recreational vehicles were allowed to discharge to the wastewater system in the previous 12 months.
- d. Analysis of data by a California licensed professional is required at least annually.

SURFACE WATER MONITORING

Because of the difficulty in monitoring bacteria in surface water, sample collection procedures must be described in a *Sampling and Analysis Plan*. Natural bacteria levels can vary significantly, and may be correlated with rainfall. When possible, surface water bacteria samples should be collected under dry weather conditions. It is critical when monitoring bacteria that all containers and surfaces a sample contacts are sterile. Sample containers must be autoclaved or manufactured to maintain sterility; use of screw top bottles, Whirl-pak[®] bags, or similar containers is acceptable. The sample hold time for bacteria samples is typically no more than six hours. Monitoring shall include, at a minimum, the following:

2. An evaluation of the performance of the wastewater treatment facility, including discussion of capacity issues, nuisance conditions, system problems, and a forecast of the flows anticipated in the next year. A flow rate evaluation as described in the General Order (Provision E.2.c) shall also be submitted.
3. If disinfection with ultraviolet light is performed, describe disinfection system maintenance activities performed in the calendar year. The description shall address inspections performed, lamp bulb replacement, lamp sleeve cleaning, and manufacturer recommended maintenance activities.
4. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the NOA and/or General Order.
5. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
6. The name and contact information for the wastewater operator responsible for operation, maintenance, and system monitoring.
7. A groundwater monitoring report prepared by a California licensed professional. This report may be prepared separately from the rest of the Annual Report. The report shall contain an analysis of groundwater data collected during the year. The analysis shall include a description of the sample events, copies of the field logs, purge method and volume, groundwater elevation and trend, a groundwater elevation map for each sample event, summary tables showing results for parameters measured, comparison of groundwater quality parameters to standards in the NOA, chain-of-custody forms, calibration logs for field equipment used, and a general evaluation of any impacts the wastewater discharge is having on groundwater quality.

A letter transmitting the monitoring reports shall accompany each report. The letter shall report violations found during the reporting period, and actions taken or planned to correct the violations and prevent future violations. The transmittal letter shall contain the following penalty of perjury statement and shall be signed by the Discharger or the Discharger's authorized agent:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of the those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

The Discharger shall implement the above monitoring program as of the date of this MRP.

Ordered by:

NAME, Executive Officer

DATE

September 23, 2014